
Big Game: Goldman Sach's Elephant Hunt in Libya

Part 2: Option Pricing



Roadmap

- Estimating volatility
- Binomial tree valuation
 - No dividends
 - Dividends
 - Floating Lookback Strike
- Black-Scholes valuation
- The LIA vs. Goldman Sachs

Estimating Volatility

Estimating Volatility

- Option pricing models require volatility as an input
- Volatility = Standard deviation of the returns of the underlying stock
- Two approaches to estimating volatility:
 1. Direct estimate from historical data
 2. Use observed prices of market-traded options on the Citigroup's stock to back out "implied volatility"... but this requires using a model (hence, more on this later, after we have a model)

Citigroup's Volatility Estimate

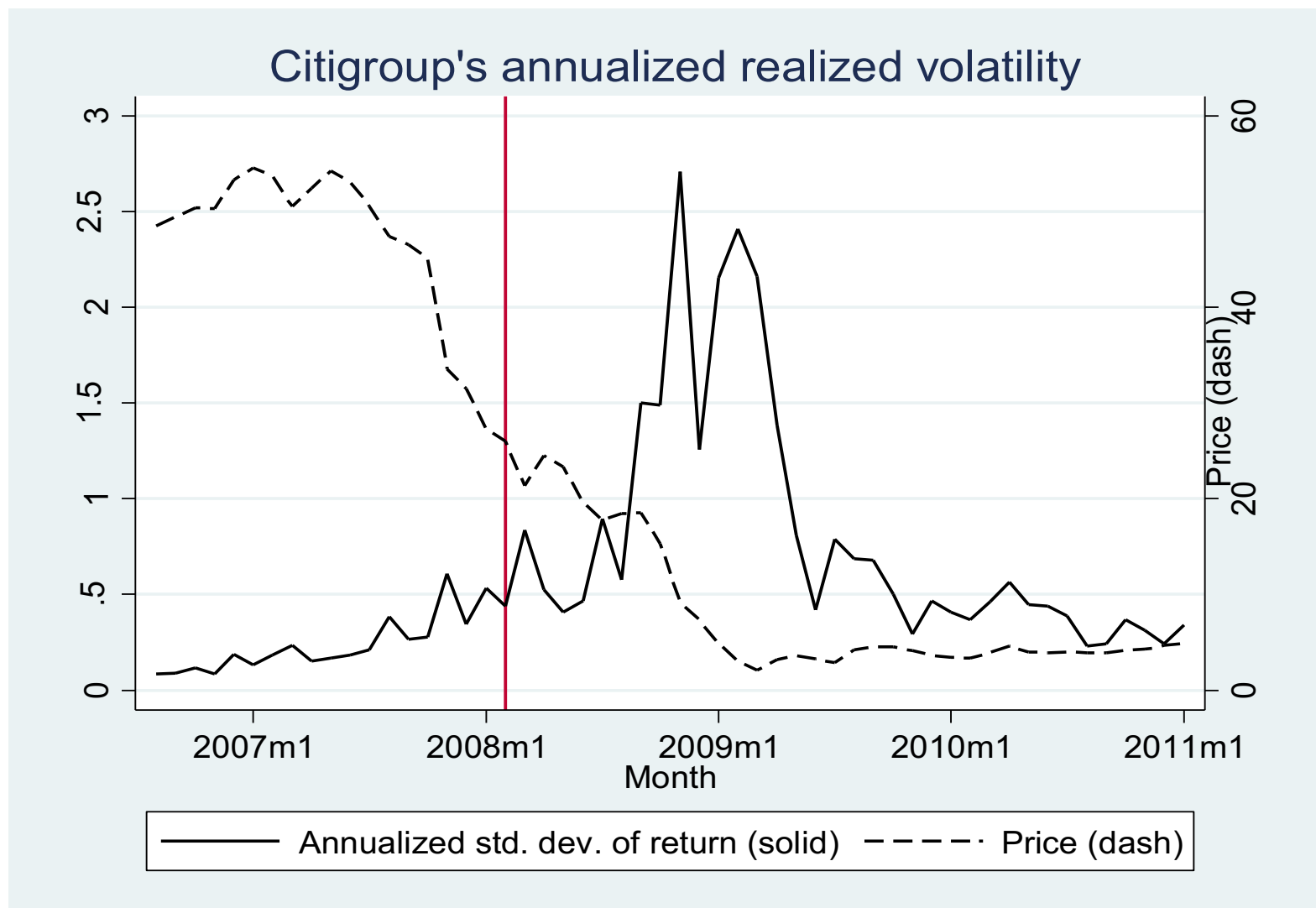


Std. Dev. Daily Return	
6 Month (July 07–Jan 08)	2.60%
12 Month (Jan 07–Jan 08)	2.04%

Multiply by $\sqrt{254}$
(# of trading days in a year)

Annualized Volatility	
6 Months	41%
12 Months	32%
Average	37%

Realized Volatility

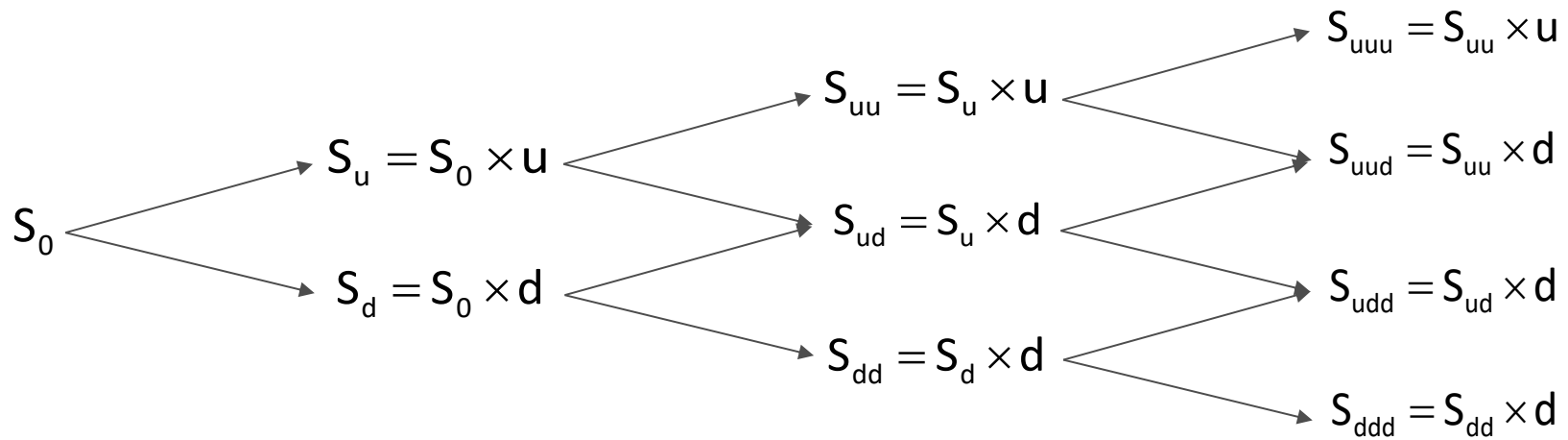


Binomial Tree: No Dividends

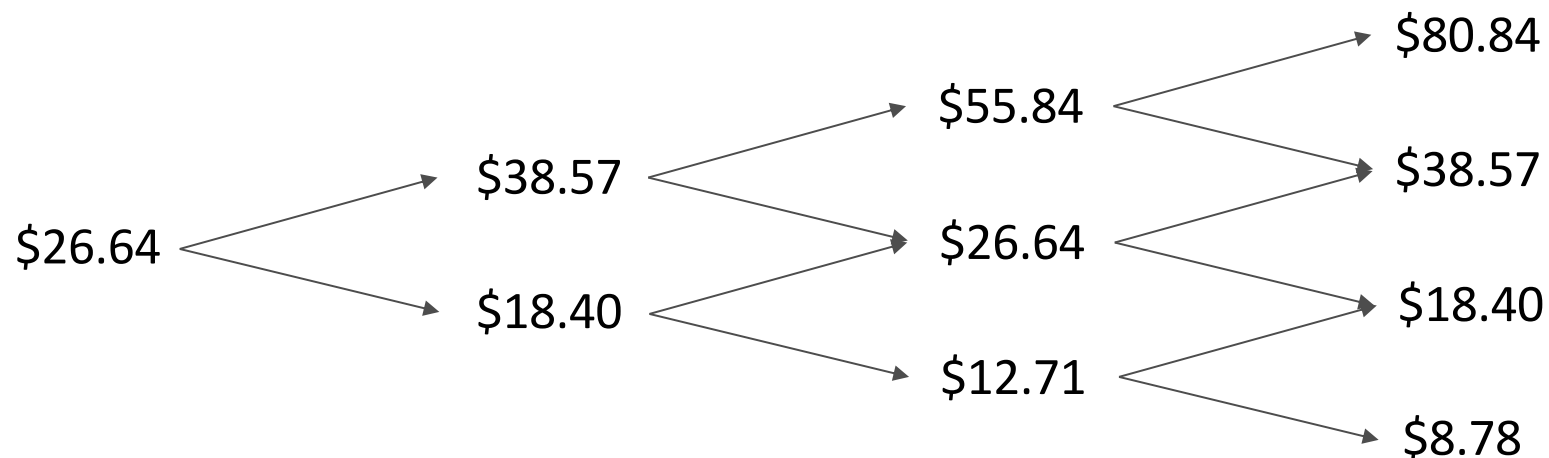
First Citigroup Trade: Parameters

Parameters	Value	Notation	Expression
Stock price	\$26.64	S_0	
Strike price	\$26.64	K	
Time to maturity	3 years	T	
Volatility	37% / year	σ	
Up-factor	1.45 / year	u	$u = e^{\sigma\sqrt{T/n}}$
Down-factor	0.69 / year	d	$d = 1/u$
Risk-free rate	2.28% / year	r	
Duration of a step	1 year	n	
Risk-neutral probability	44%	p	$p = \frac{(1+r) - d}{u - d}$

Stock Price Tree (w/o Dividends)



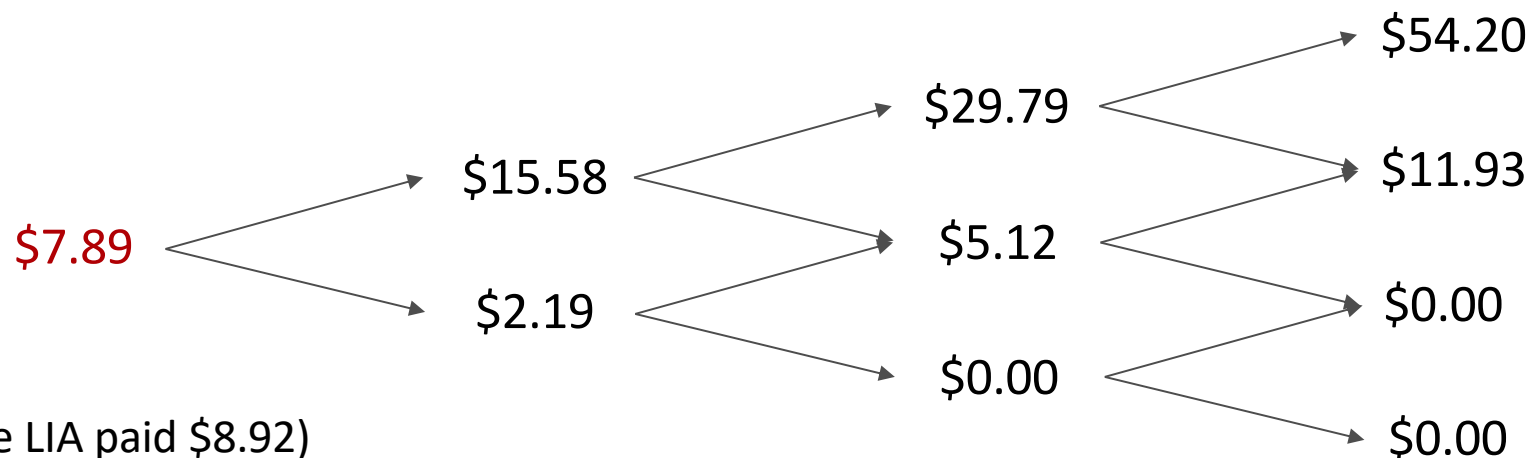
Build stock tree forwards



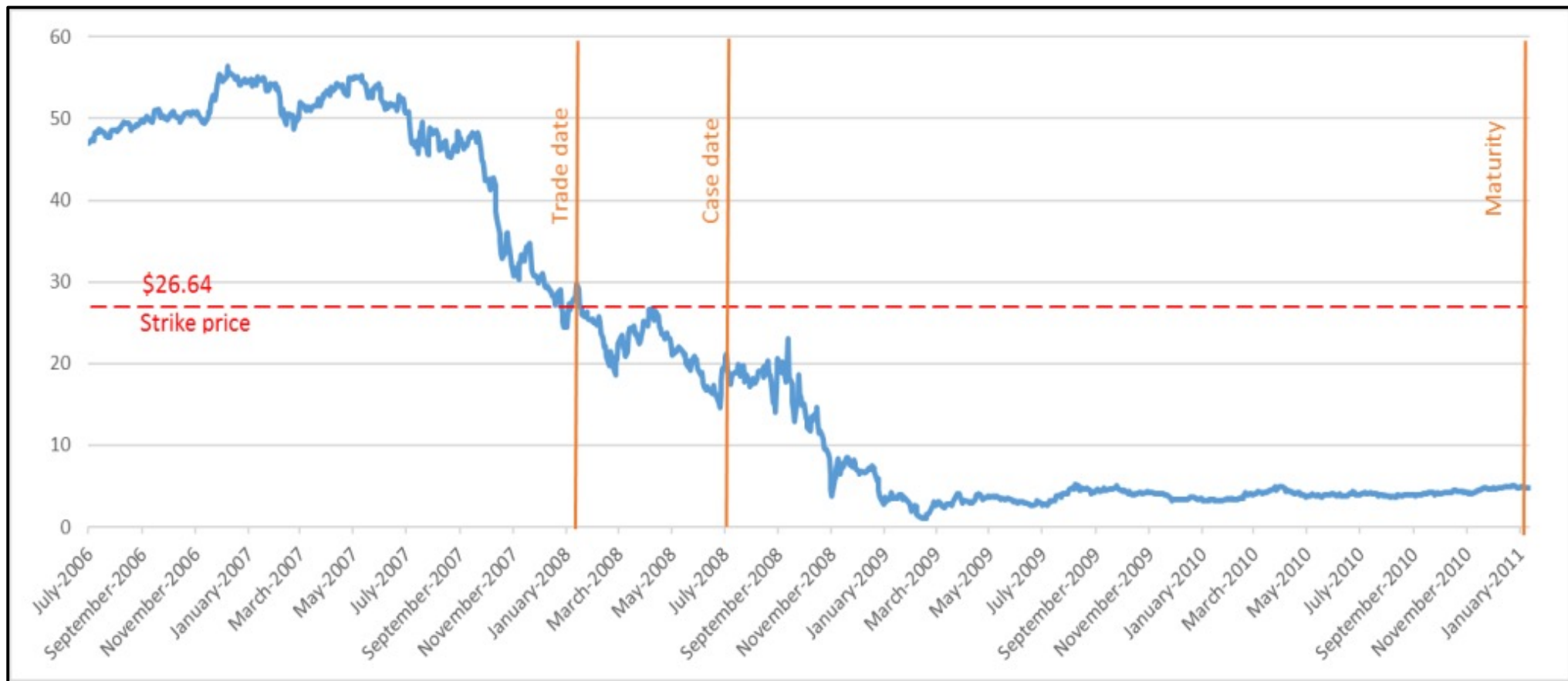
Call Price Tree (w/o Dividends)

$$\begin{array}{lcl}
 \dots & C_{uu} = \frac{p \times C_{uuu} + (1-p) \times C_{uud}}{1+r} & \begin{array}{l} C_{uuu} = \text{Max}(S_{uuu} - K, 0) \\ C_{uud} = \text{Max}(S_{uud} - K, 0) \end{array} \\
 & C_{ud} = \frac{p \times C_{udd} + (1-p) \times C_{udd}}{1+r} & \begin{array}{l} C_{udd} = \text{Max}(S_{udd} - K, 0) \\ C_{udd} = \text{Max}(S_{udd} - K, 0) \end{array} \\
 \dots & C_{dd} = \frac{p \times C_{ddd} + (1-p) \times C_{ddd}}{1+r} & C_{ddd} = \text{Max}(S_{ddd} - K, 0)
 \end{array}$$

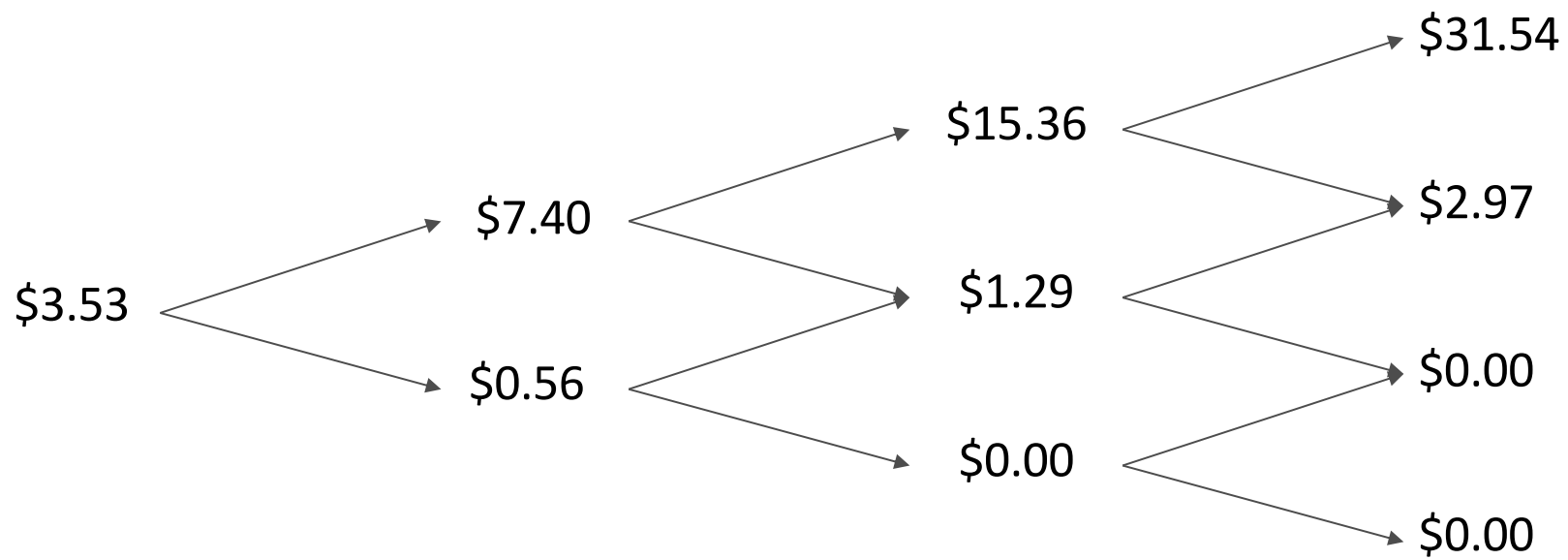
Build option tree backwards



Citigroup's Stock Price Evolution

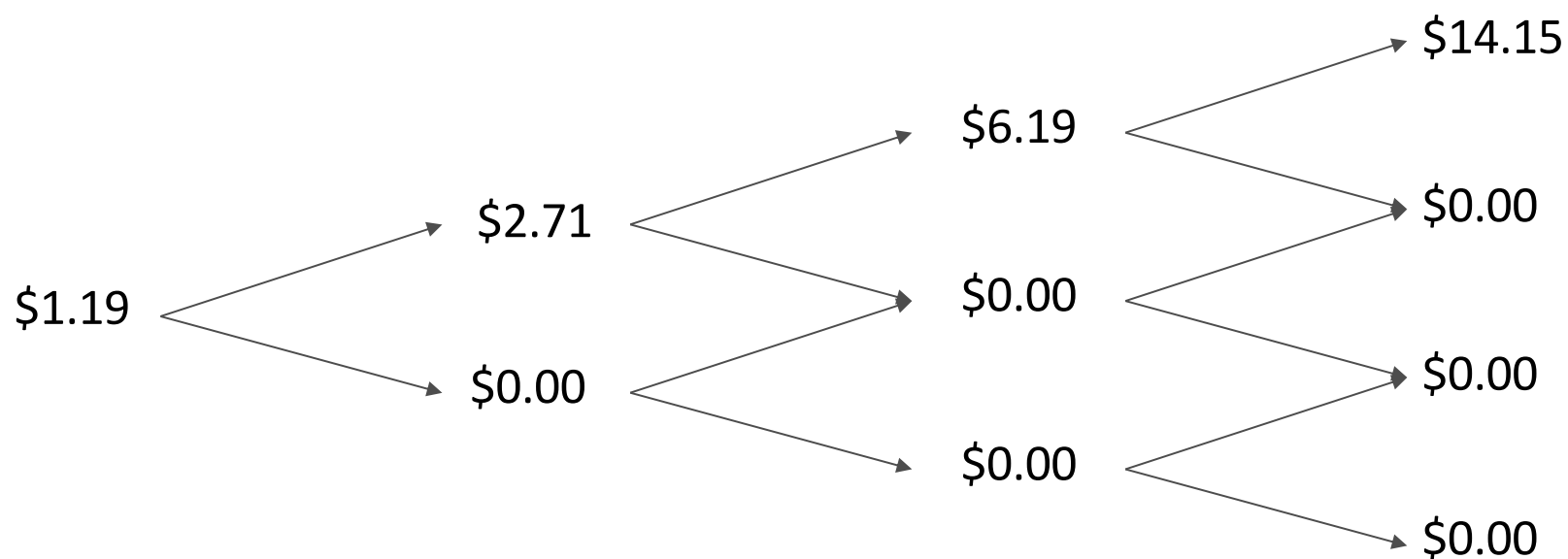


Case Time: July 23, 2008



Uses the 25/1/2008 vol. (37%)

Post-Lehman Crash: Sept. 15, 2008

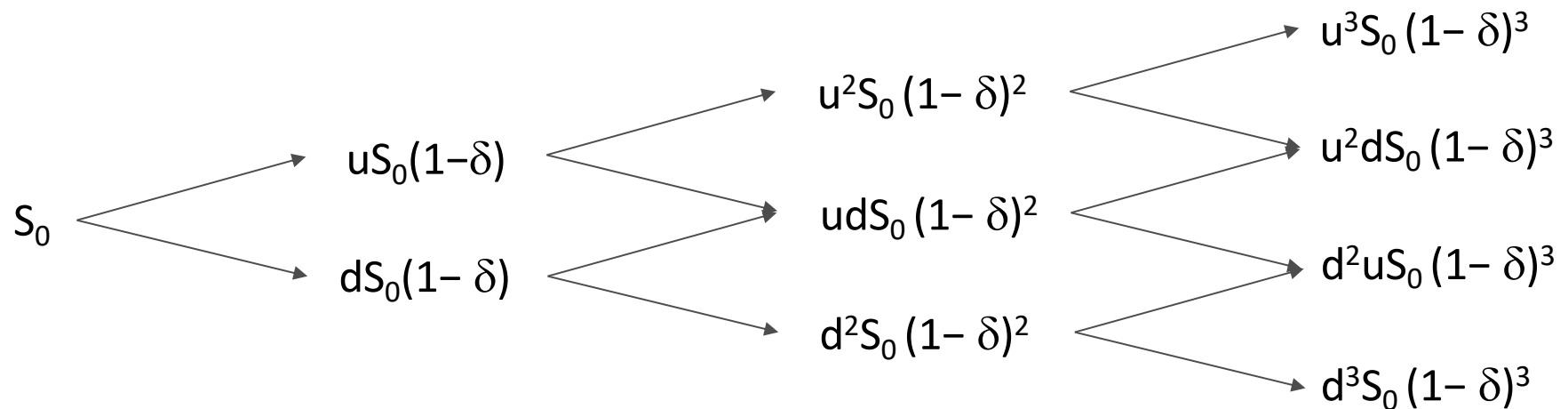


Volatility of 37%

Binomial Tree: Dividends

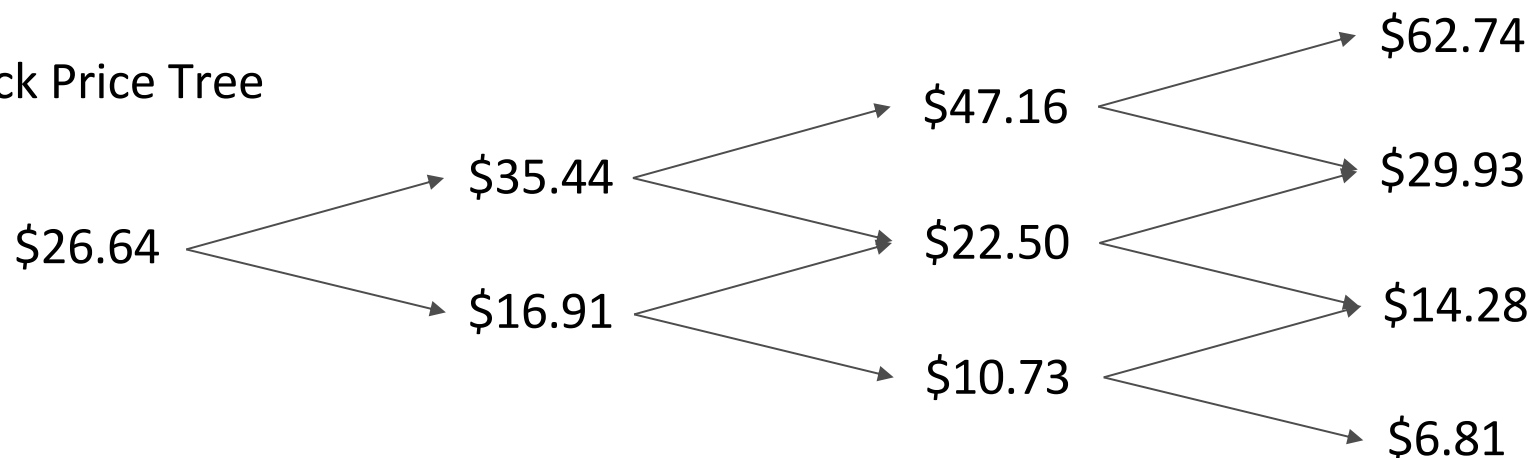
Stock Price Tree with Dividends

- Assume the stock pays a constant (annual) dividend yield δ
- Reduce the stock price at each node by a factor $(1-\delta)$
- The call price tree is built the same way as before

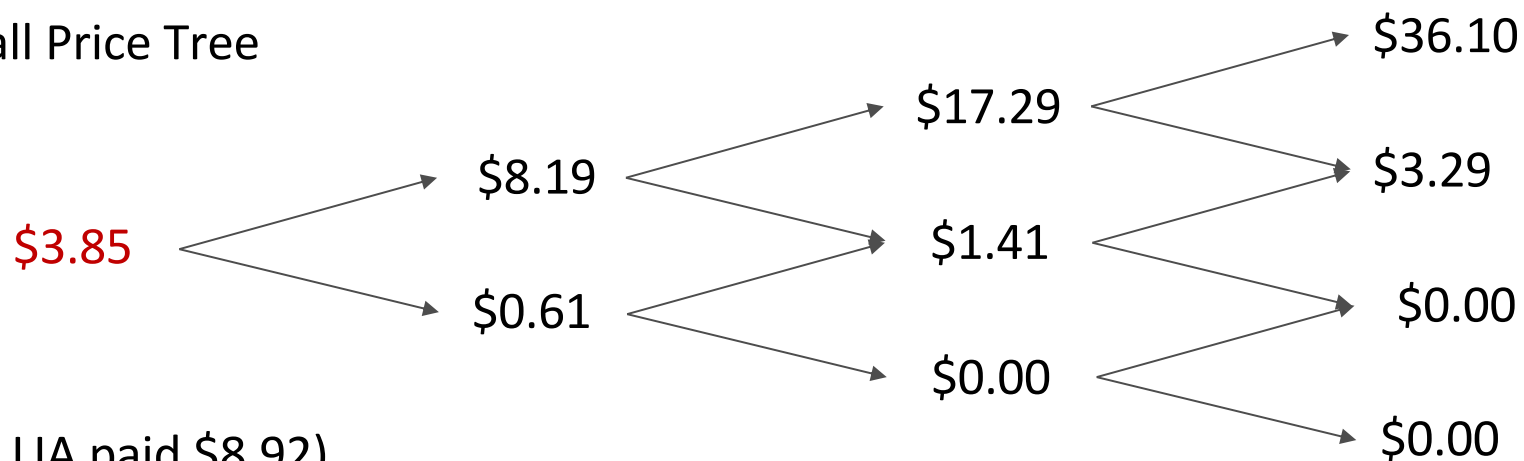


Trees (Historical $\delta = 8.1\%$)

- Stock Price Tree



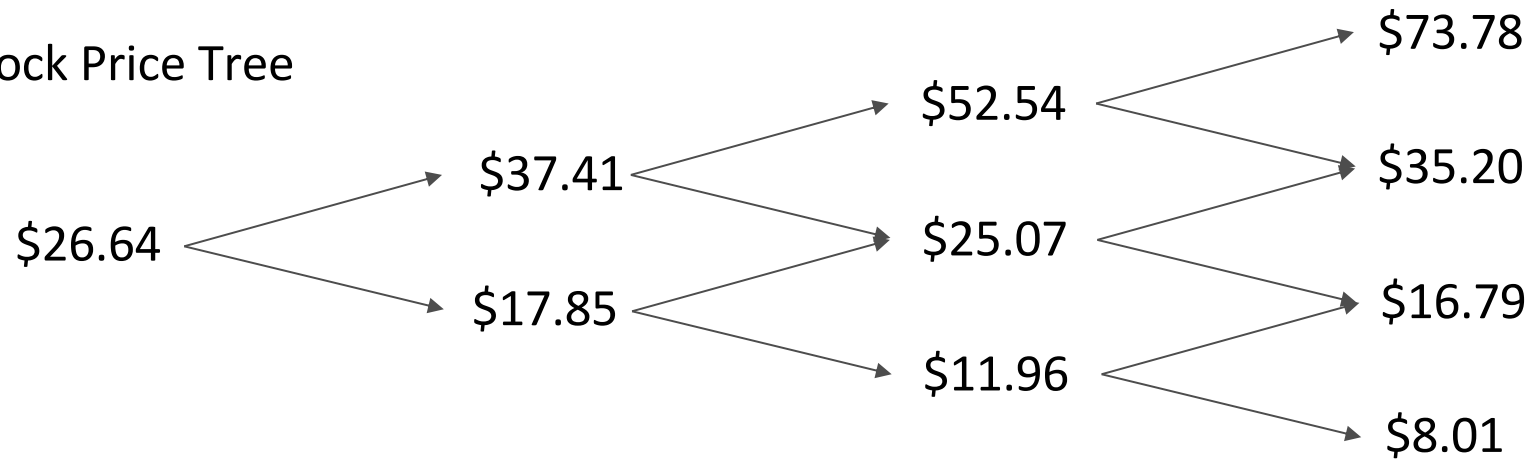
- Call Price Tree



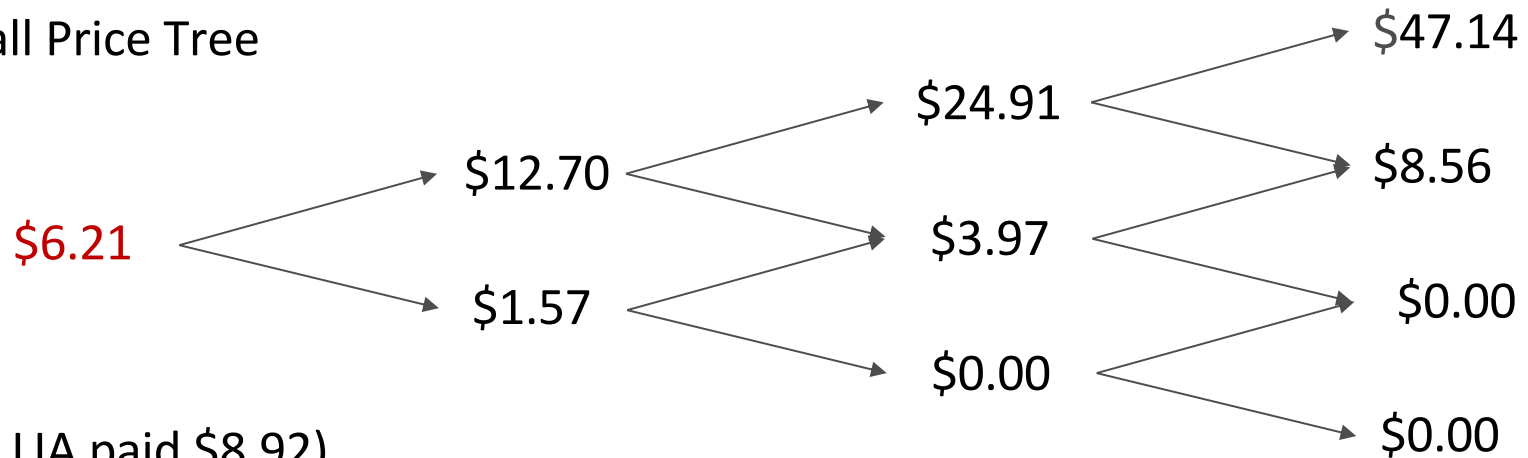
(The LIA paid \$8.92)

Trees ($\delta = 3\%$)

- Stock Price Tree

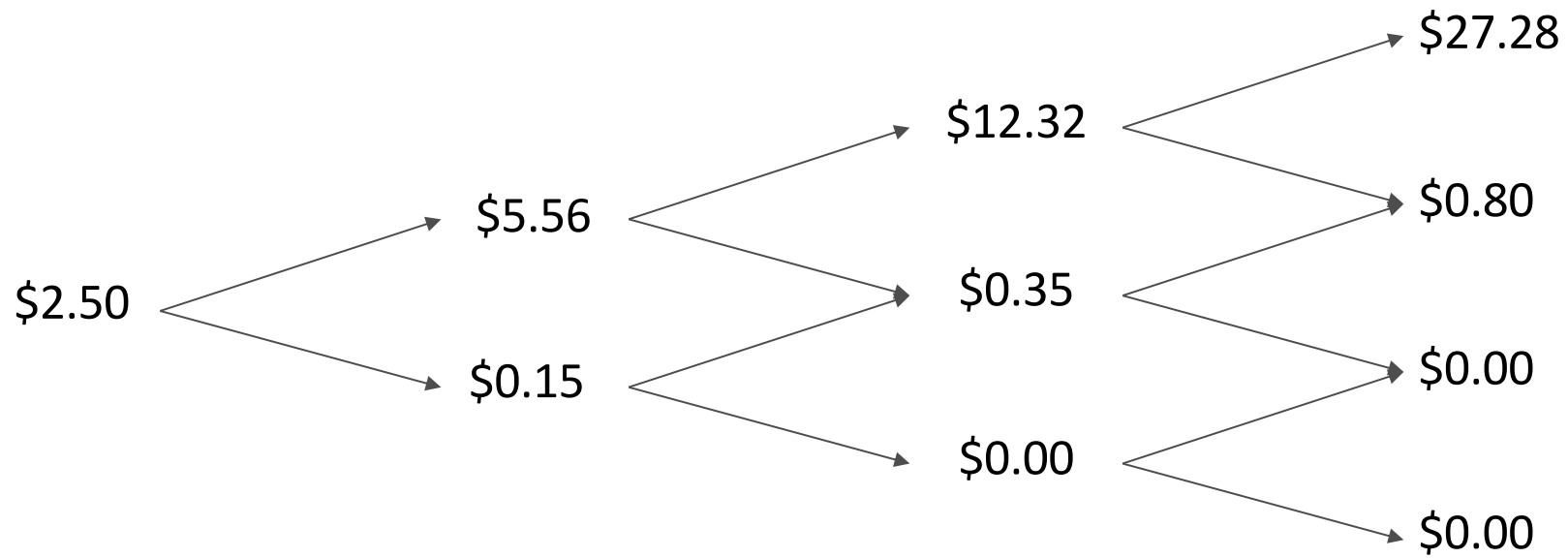


- Call Price Tree



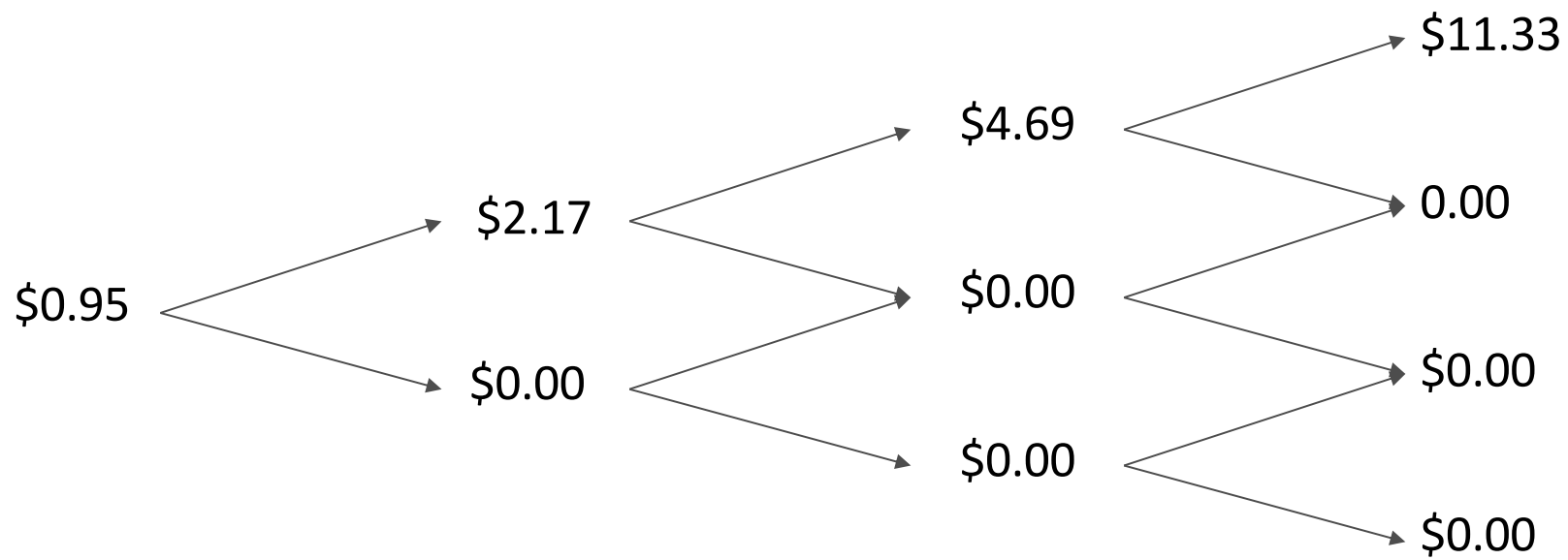
(The LIA paid \$8.92)

Case Time: July 23, 2008 ($\delta = 3\%$)



Uses the 25/1/2008 vol. (37%)

Post-Lehman Crash: Sept. 15, 2008 ($\delta = 3\%$)



Volatility of 37%

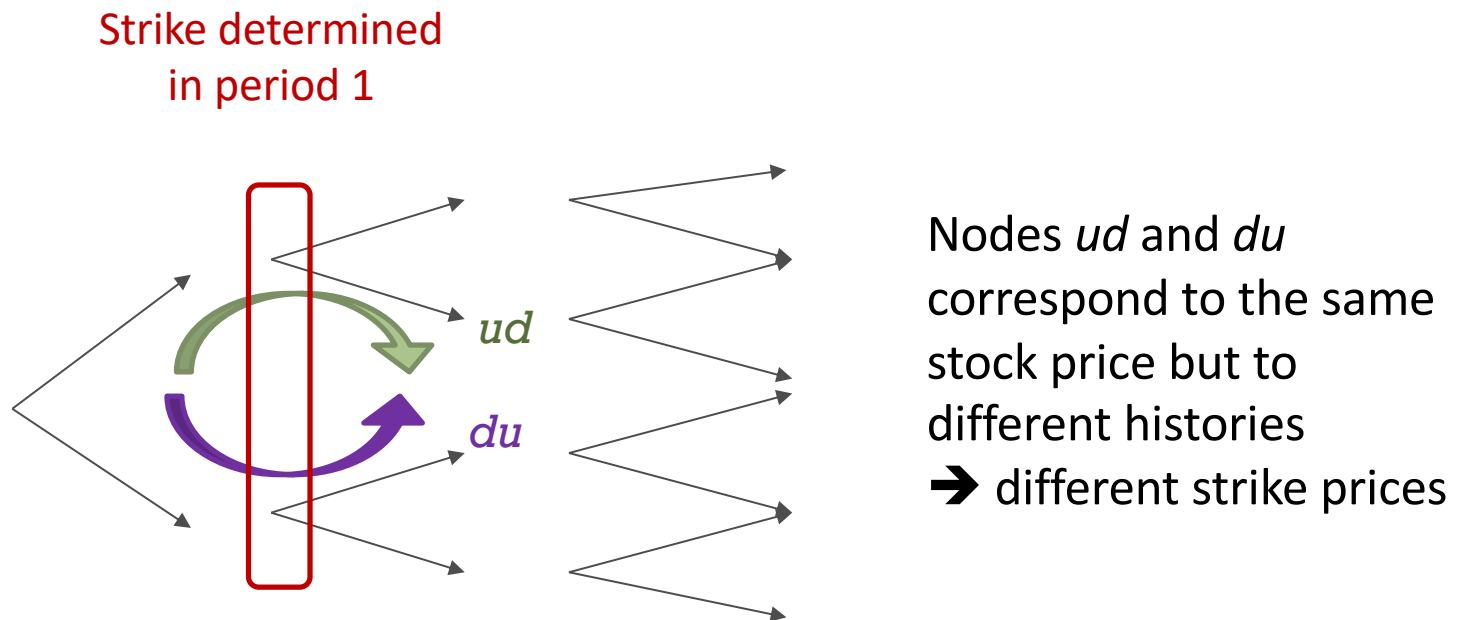
Binomial Tree: Floating Lookback Strike

Floating Lookback Strike

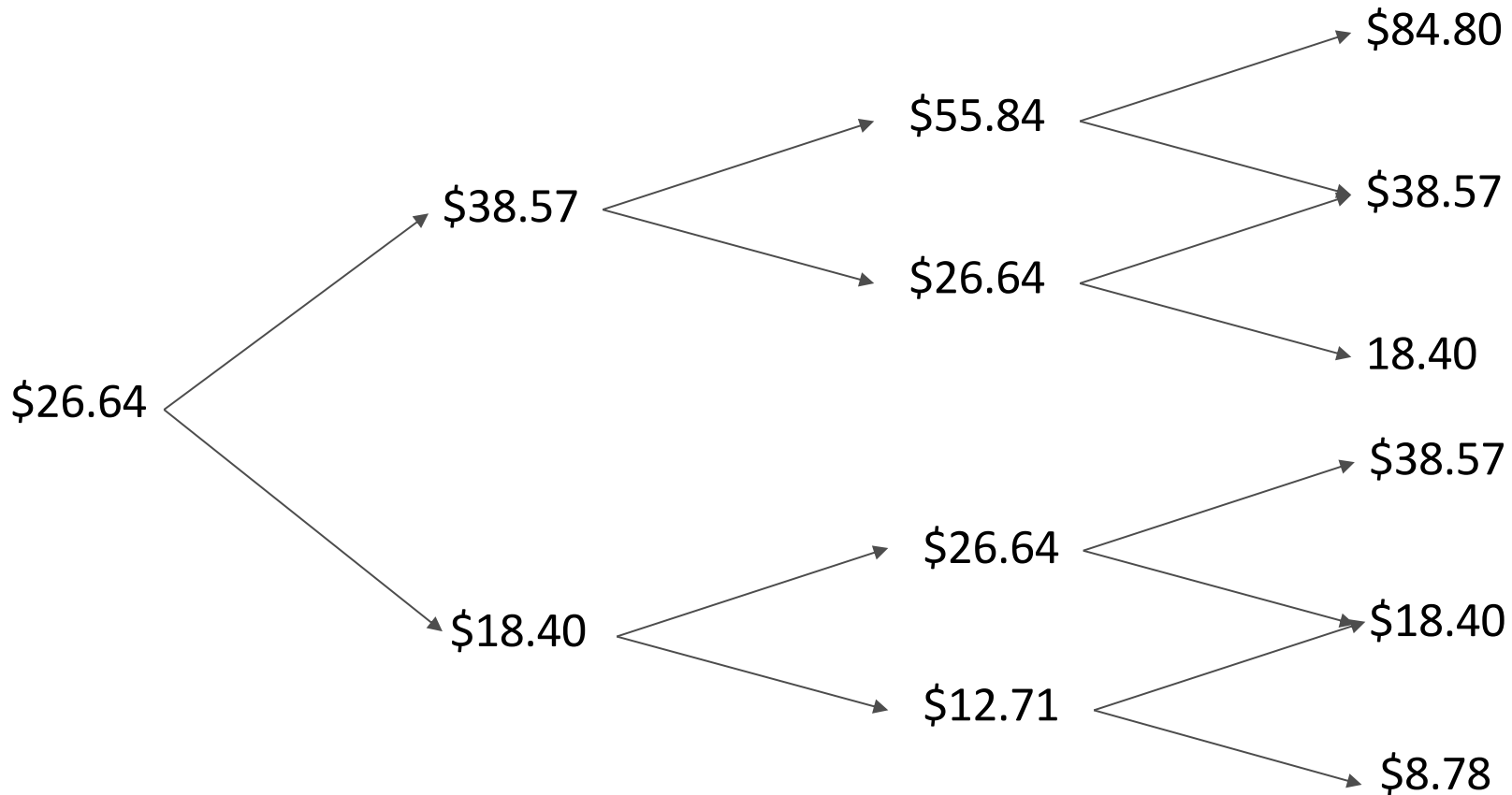
- Strike = lower of
 1. Current stock price = \$26.64
 2. 9 monthly average stock prices in first 9 months (floating period)
 - All subject to a floor of $90\% \times \$26.64 = \23.98
 - Simplifying assumptions:
 - 12-month floating period (rather than 9 months)
 - Use the price at the end of the floating period in lieu of the lowest of 9 monthly average prices
- ➔ Strike price = $\max\{\min(S_0, S_1), 90\% \times S_0\}$ where S_0 and S_1 are stock prices at inception and after 1 year

Valuation with Floating Lookback Strike

- Different paths (histories) followed by the stock price produce different strikes
 - ➔ Lookback makes the option's payoff path-dependent
 - ➔ Binomial trees need to bifurcate
 - + Though an '*up* followed by a *down*' (*ud*) produces the same stock price as a '*down* followed by an *up*' (*du*), we must now distinguish these 2 nodes
 - + From year 2 on, trees are standard (no more price-dependency)

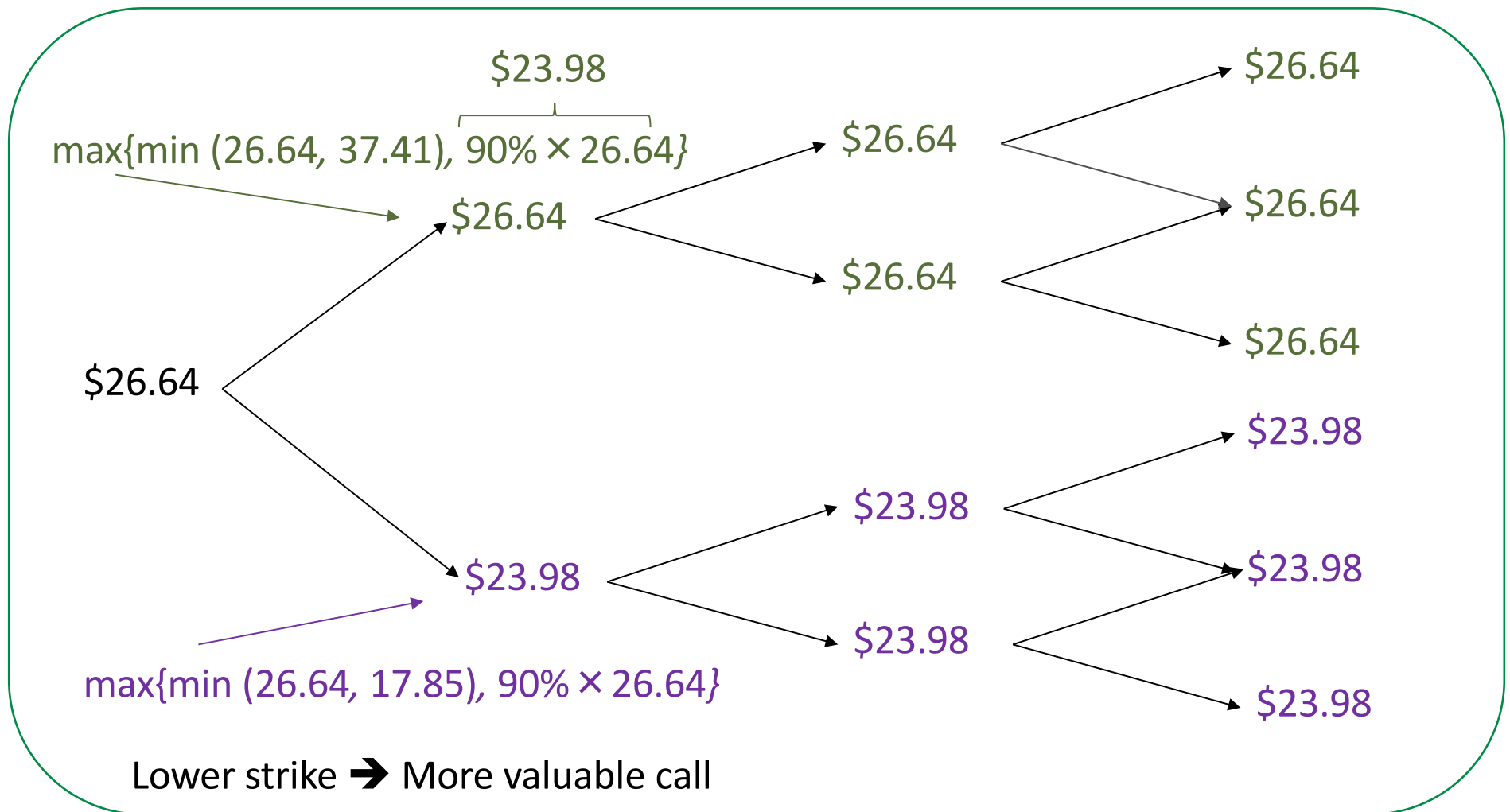


Stock Price Tree ($\delta = 0\%$)

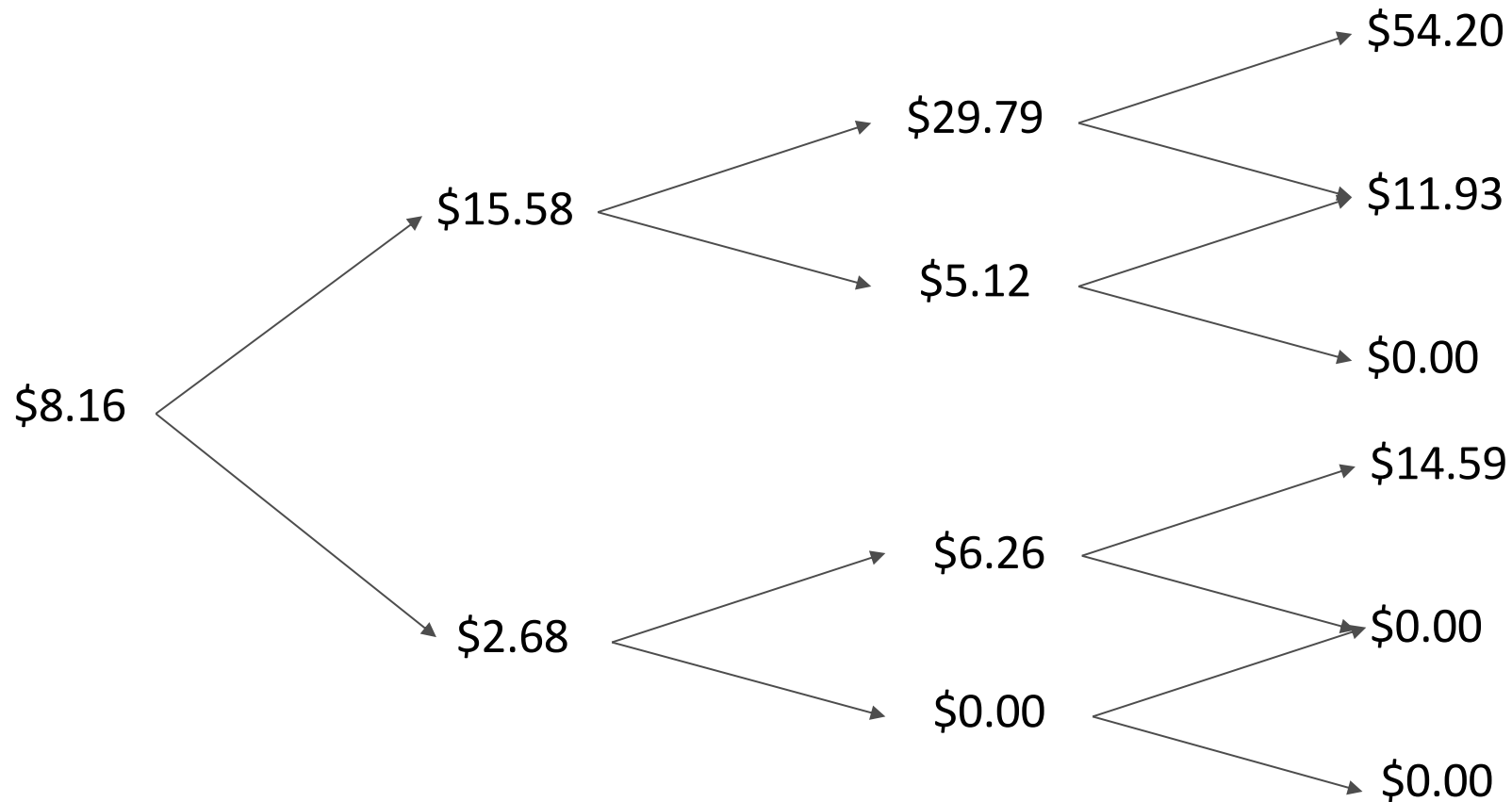


(Values no different than normal stock price tree)

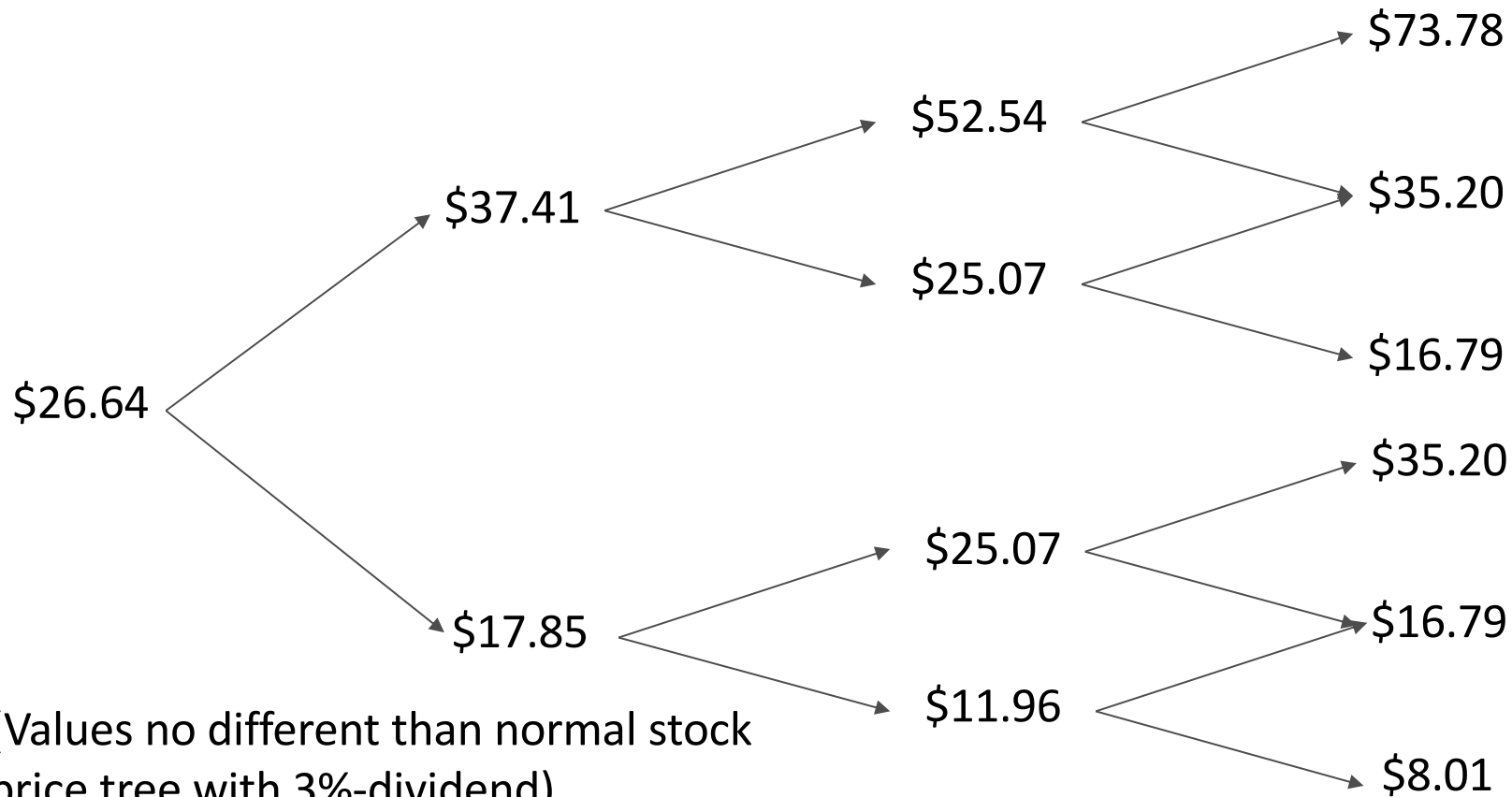
Strike Price Tree ($\delta = 0\%$)



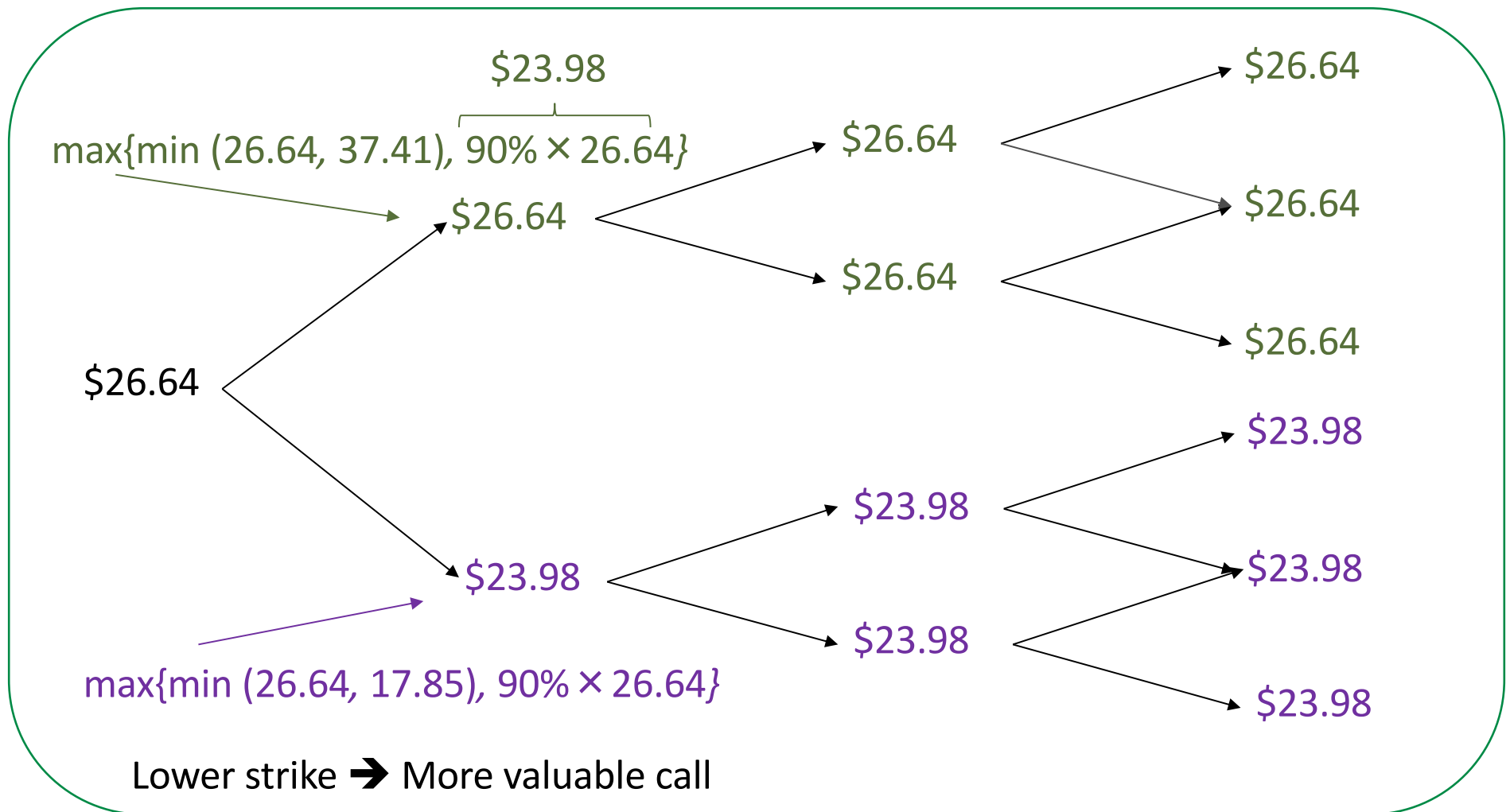
Call Price Tree ($\delta = 0\%$)



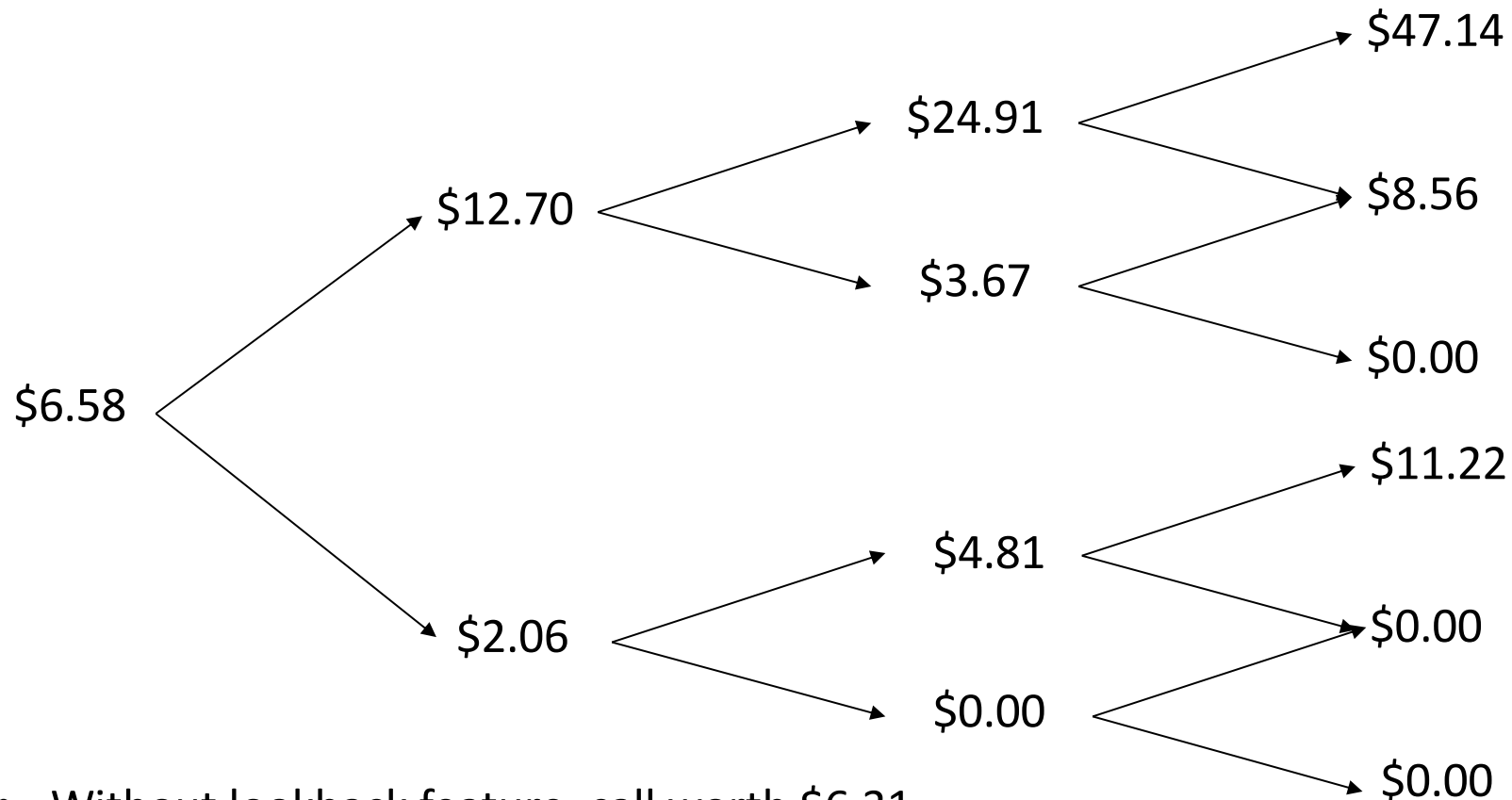
Stock Price ($\delta = 3\%$)



Strike Price ($\delta = 3\%$)



Call Price ($\delta = 3\%$)

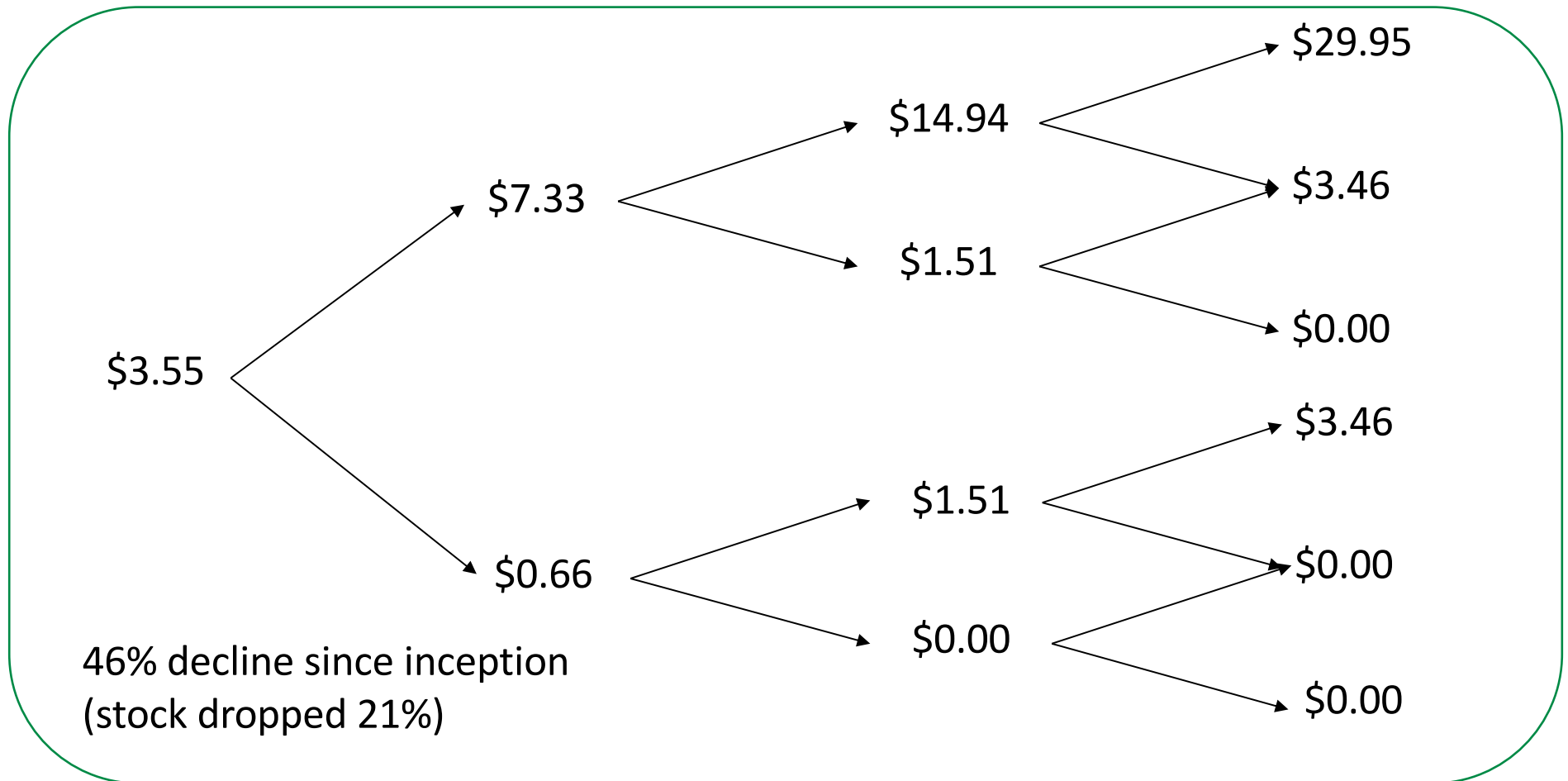


- Without lookback feature, call worth \$6.31
- LIA paid \$8.92 (26% overpriced)
- Ignoring div. (ex post correct yield), call only 9% overpriced

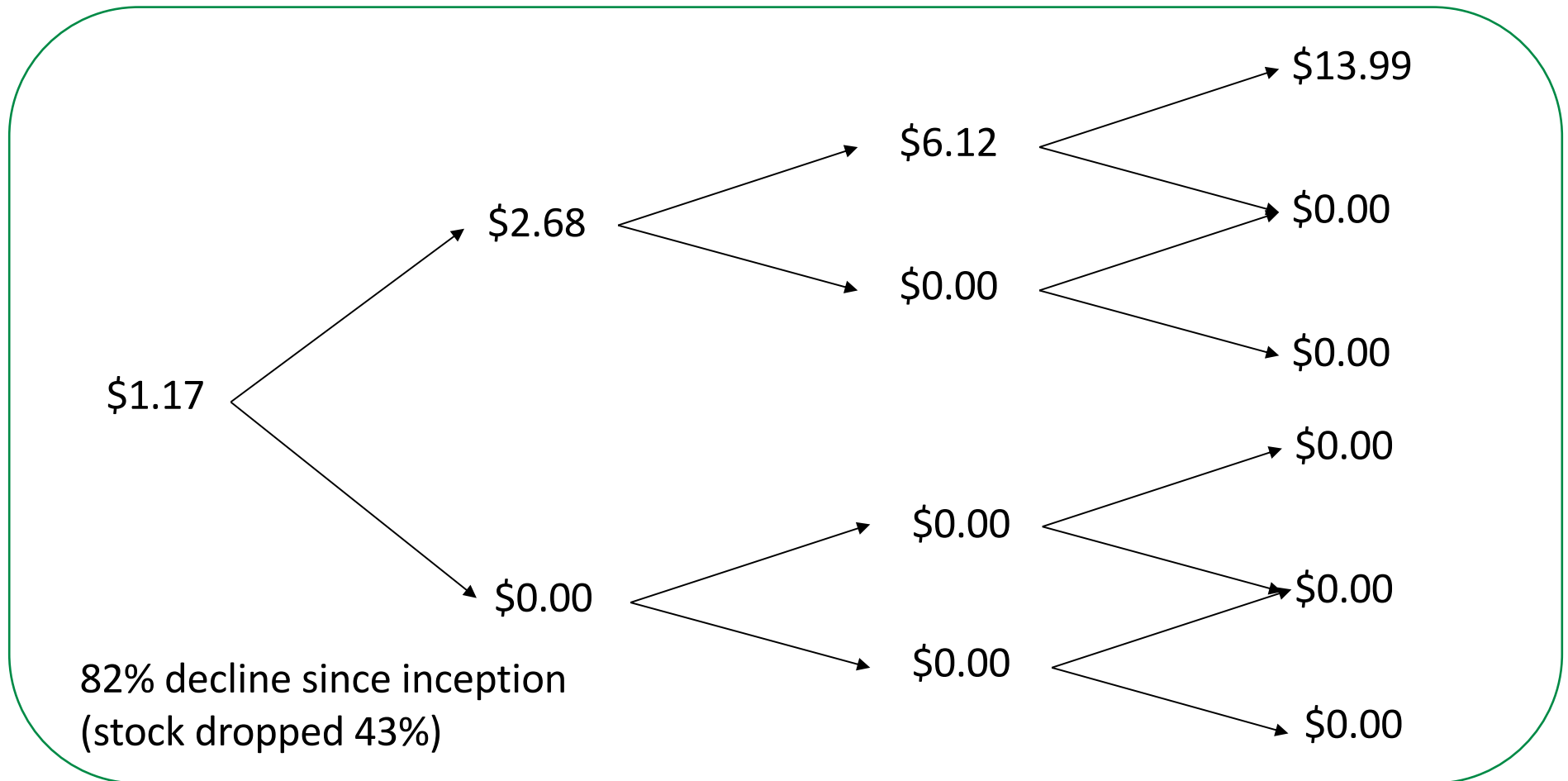
Valuation Summary

AT INCEPTION - JANUARY 25, 2008	Binomial Tree	
	Call (\$)	Trade (\$m)
Standard call without dividends	7.89	88.3
Standard call with dividend yield of 3%	6.31	70.7
Standard call with dividend yield of 8.1%	3.85	43.1
Floating lookback without dividends	8.16	91.4
Floating lookback with dividend yield of 3%	6.58	73.7
Floating lookback with dividend yield of 8.1%	4.11	46.1

Case Time: July 23, 2008 ($\delta = 3\%$)



Post-Lehman Crash: Sept. 15, 2008 ($\delta = 3\%$)



Black-Scholes Formula

Black-Scholes Formula (No Dividends)

$$C = N(d_1) \times S - N(d_2) \times PV(K)$$

with $d_1 = \frac{\ln(S/PV(K)) + 0.5 \times \sigma^2 T}{\sigma \sqrt{T}}$ and $d_2 = d_1 - \sigma \sqrt{T}$

- $N(d)$ is the value of at d of the CDF of the standard normal distribution

$$N(d) = \text{NORMDIST}(d; 0; 1; 1)$$

- S is the current price of the underlying asset
- $PV(K)$ is the present value of strike price K using the risk-free rate, annually or continuously compounded
- T is the time to maturity
- σ is the volatility of stock returns

Value Evolution

CALL VALUE AT INCEPTION (JAN. 25 2008)	
Inputs	
Stock price (S)	26.64
Strike price (K)	26.64
Volatility (annualized) (σ)	37%
Risk-free rate (r)	2.28%
Time to maturity (years)	3.00
Black-Scholes formula	
d_1	0.43
d_2	-0.21
$N(d_1)$	0.66
$N(d_2)$	0.41
Call value	7.38
Trade (\$m)	82.7

CALL VALUE CASE TIME (JULY 28, 2008)	
Inputs	
Stock price (S)	21.12
Strike price (K)	26.64
Volatility (annualized) (σ)	37%
Risk-free rate (r)	2.28%
Time to maturity (years)	2.50
Black-Scholes formula	
d_1	-0.01
d_2	-0.59
$N(d_1)$	0.50
$N(d_2)$	0.28
Call value	3.53
Trade (\$m)	39.5

CALL VALUE POST-LEHMAN (SEPT. 15, 2008)	
Inputs	
Stock price (S)	15.24
Strike price (K)	26.64
Volatility (annualized) (σ)	37%
Risk-free rate (r)	2.28%
Time to maturity (years)	2.375
Black-Scholes formula	
d_1	-0.60
d_2	-1.17
$N(d_1)$	0.27
$N(d_2)$	0.12
Call value	1.13
Trade (\$m)	12.6

Black-Scholes Formula with Dividends

$$C = N(d_1) \times S(1 + \delta)^{-T} - N(d_2) \times PV(K)$$

$$\text{with } d_1 = \frac{\ln(S(1 + \delta)^{-T} / PV(K)) + 0.5 \times \sigma^2 T}{\sigma \sqrt{T}} \quad \text{and } d_2 = d_1 - \sigma \sqrt{T}$$

- S is the current price of the underlying asset
- $PV(K)$ is the present value of strike price K using the risk-free rate, annually or continuously compounded
- δ is the annually compounded dividend yield
- T is the time to maturity
- σ is the volatility of stock returns

Value Evolution

CALL VALUE AT INCEPTION (JAN. 25 2008)	
Inputs	
Stock price (S)	26.64
Strike price (K)	26.64
Volatility (annualized) (σ)	37%
Risk-free rate (r)	2.28%
Time to maturity (years)	3.00
Dividend yield (δ)	3.00%
Black-Scholes formula	
d_1	0.29
d_2	-0.35
$N(d_1)$	0.61
$N(d_2)$	0.36
Call value	5.94
Trade (\$m)	66.5

CALL VALUE CASE TIME (JULY 28, 2008)	
Inputs	
Stock price (S)	21.12
Strike price (K)	26.64
Volatility (annualized) (σ)	37%
Risk-free rate (r)	2.28%
Time to maturity (years)	2.50
Dividend yield (δ)	3.00%
Black-Scholes formula	
d_1	-0.13
d_2	-0.72
$N(d_1)$	0.45
$N(d_2)$	0.24
Call value	2.82
Trade (\$m)	31.6

CALL VALUE POST-LEHMAN (SEPT. 15, 2008)	
Inputs	
Stock price (S)	15.24
Strike price (K)	26.64
Volatility (annualized) (σ)	37%
Risk-free rate (r)	2.28%
Time to maturity (years)	2.375
Dividend yield (δ)	3.00%
Black-Scholes formula	
d_1	-0.72
d_2	-1.29
$N(d_1)$	0.23
$N(d_2)$	0.10
Call value	0.86
Trade (\$m)	9.7

Conclusion

Valuation Summary

- Absent dividends:
 - All models value the trade at about \$80-\$90m
 - The floating lookback strike adds limited value (about \$3m)
- The trade's value is very sensitive to assumptions about dividends
 1. The stock price is sensitive to dividend payments
 2. The leverage effect makes option values sensitive to stock prices
- Across methods and assumptions:
 - The trade had lost a lot of value by July 2008
 - And was set to lose even more value after Lehman Brothers crashed

Value Estimate vs. \$100m

- “*Technical/Methodological Problems*”
 - Under-estimated volatility and over-estimated dividends?
 - Models are only models:
 - + Assumptions about the stock price evolution may be violated
 - + Goldman will “charge” the LIA for transaction costs and imperfect hedging (i.e. models give *mid-market price*, not *walkaway price*)
- *Market Power* → Goldman can earn a large margin
 - The LIA needs Goldman: lack of expertise + avoid sanctions
 - The market for such large tailored trades may not be very competitive, i.e., only a few banks can organize them effectively

Value Estimate vs. \$100m (continued)

- *Did Goldman exploit the LIA?*
 - The LIA had little expertise → Relied excessively on Goldman
 - Potential conflicts of interest
 - + Goldman's scenario analysis starts at \$32 (→ 20% price rise)
 - The LIA did not have sound due diligence processes
 - + Independent advisors carried little weight (Baruni resignation)
 - + McDougall (lawyer) asked to see the due diligence → “Due what?”
 - The LIA sued Goldman for having exploited its limited experience

Conclusion

- Much seems to ride of the expectation of dividends.
- Say dividend forecasts are very low (yield close to zero):
 - NB: In hindsight, Citigroup suspended its dividend in 2009 and 2010
 - ➔ Trade value $\approx$ \$90m ➔ $\approx$ 10% for costs, risk reward, and profit
- Say 3% dividend yield (a plausible expectation)
 - ➔ Trade value $\approx$ \$75m ➔ Must explain 25% of the premium
- The LIA was perhaps incensed by realizing how sensitive the trade's value was to changes in the stock price i.e. the leverage effect

What Happened?

Epilogue

- Kabbaj flees Libya
 - Sidelined + No \$9m bonus → Settlement for \$4.5m + resignation
- 2011: Libyan revolution
 - Gaddafi killed; Zarti flees Libya
 - All nine elephant trades expire worthless
- 2014: The LIA sues Goldman for “undue influence” in the 9 trades
- Oct. 2016: London High Court rules in Goldman’s favor
- May 2017: SocGen settles for €963m in suit with LIA over \$2.1bn of trades

But Love is Stronger

In 2014, Sofia Wellesley
married British pop
singer James Blunt

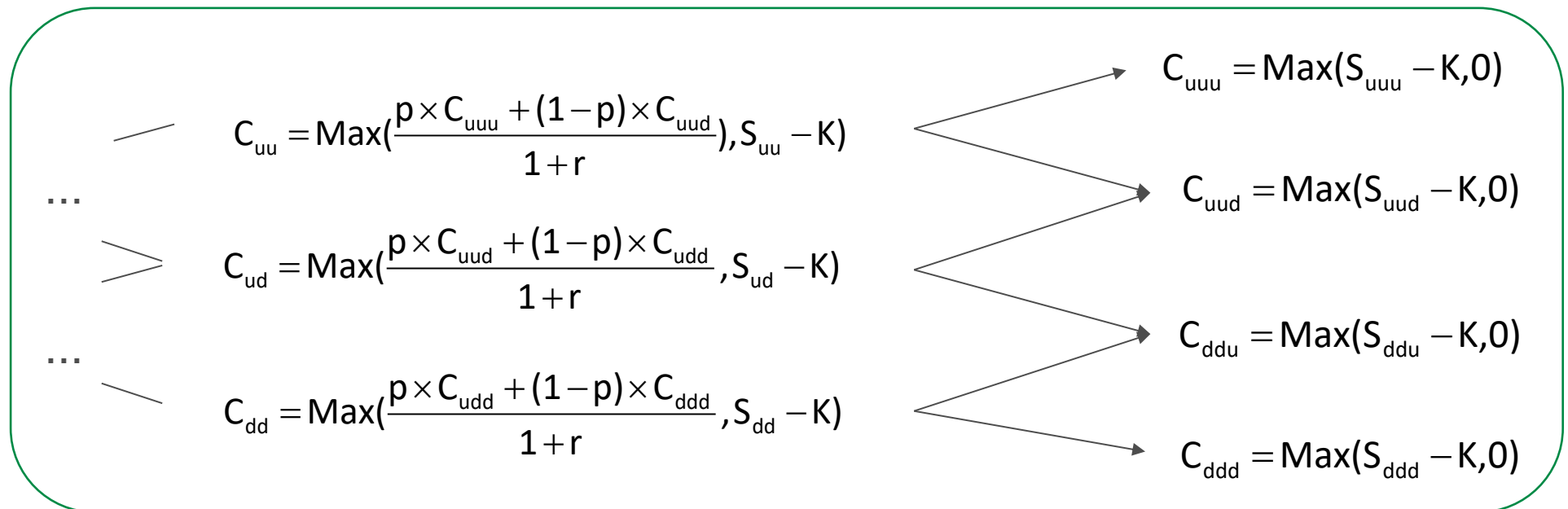


Appendix C.1:

Binomial Tree for American Options

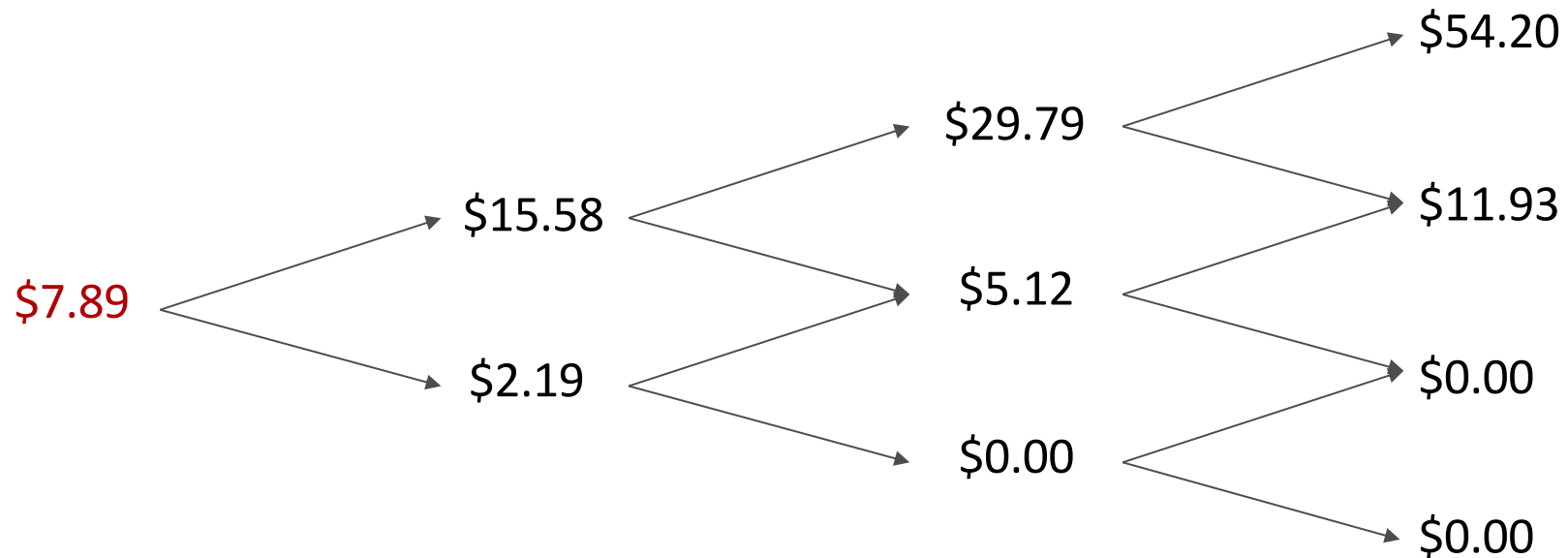
American Call Options

- For American options, we need to check at each node whether early exercise is optimal:



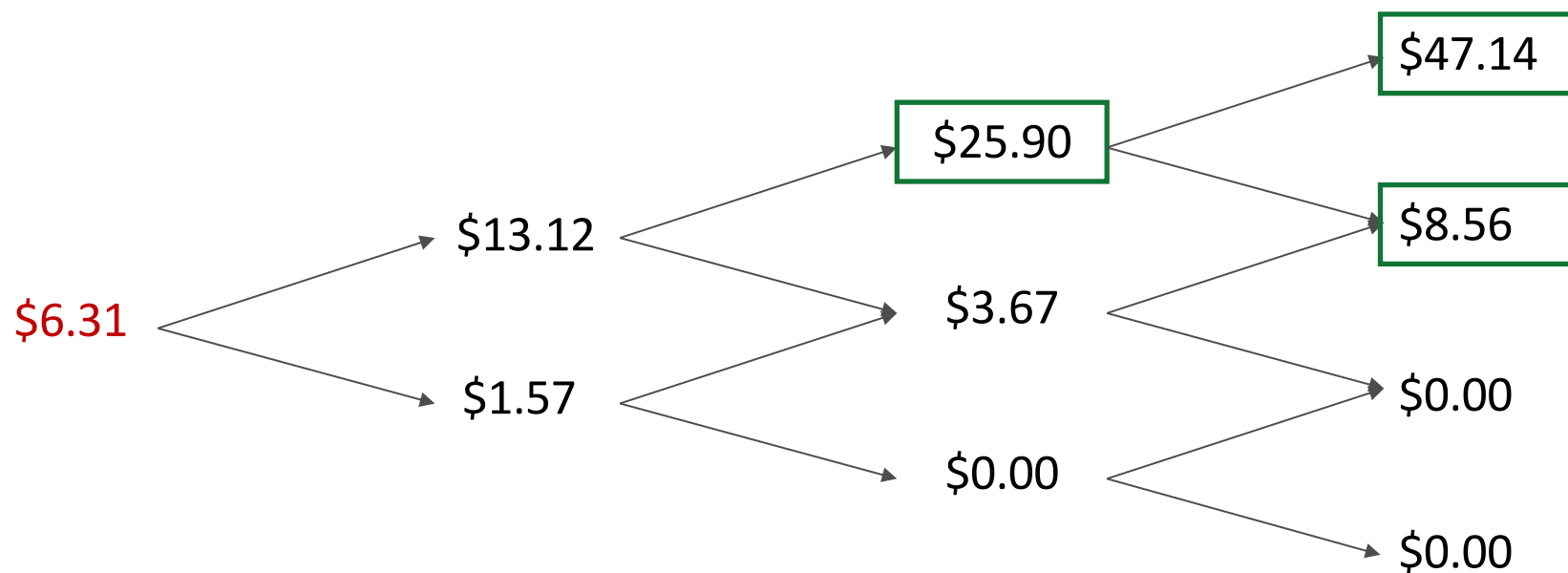
Trade as American Call ($\delta = 0\%$)

Early exercise is never optimal →
Same price as for European call



= Call will be exercised

Trade as American Call ($\delta = 3\%$)

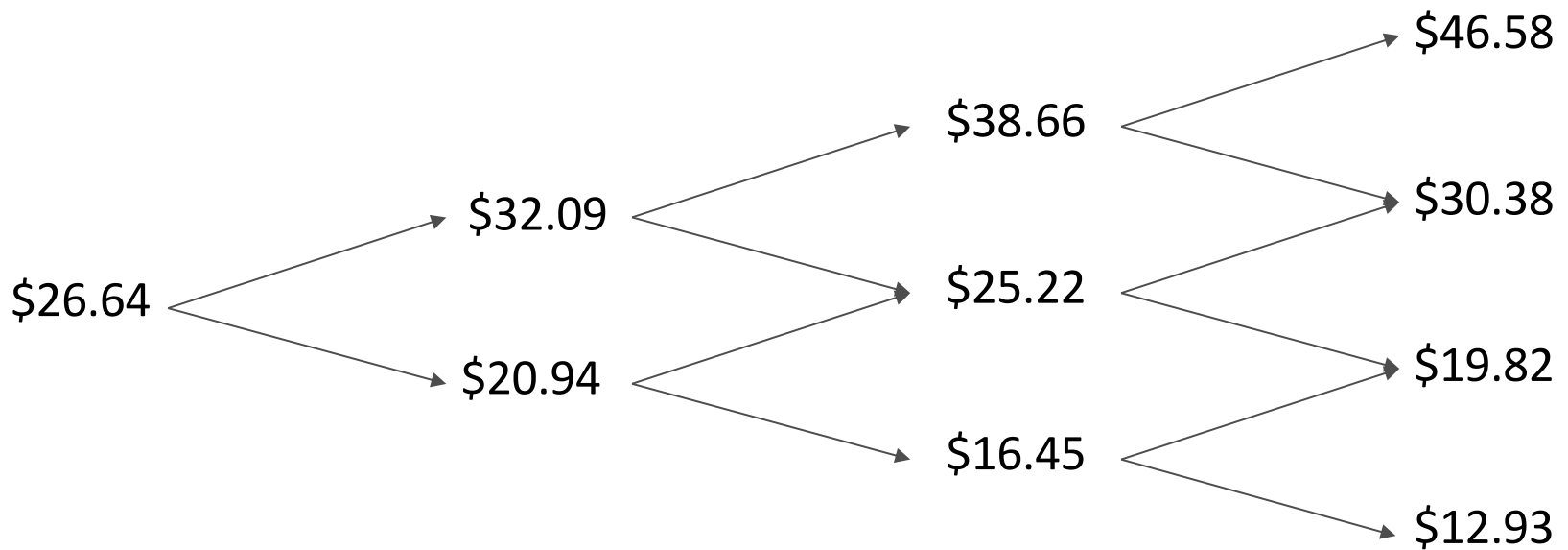


 = Call will be exercised

Appendix C.2:

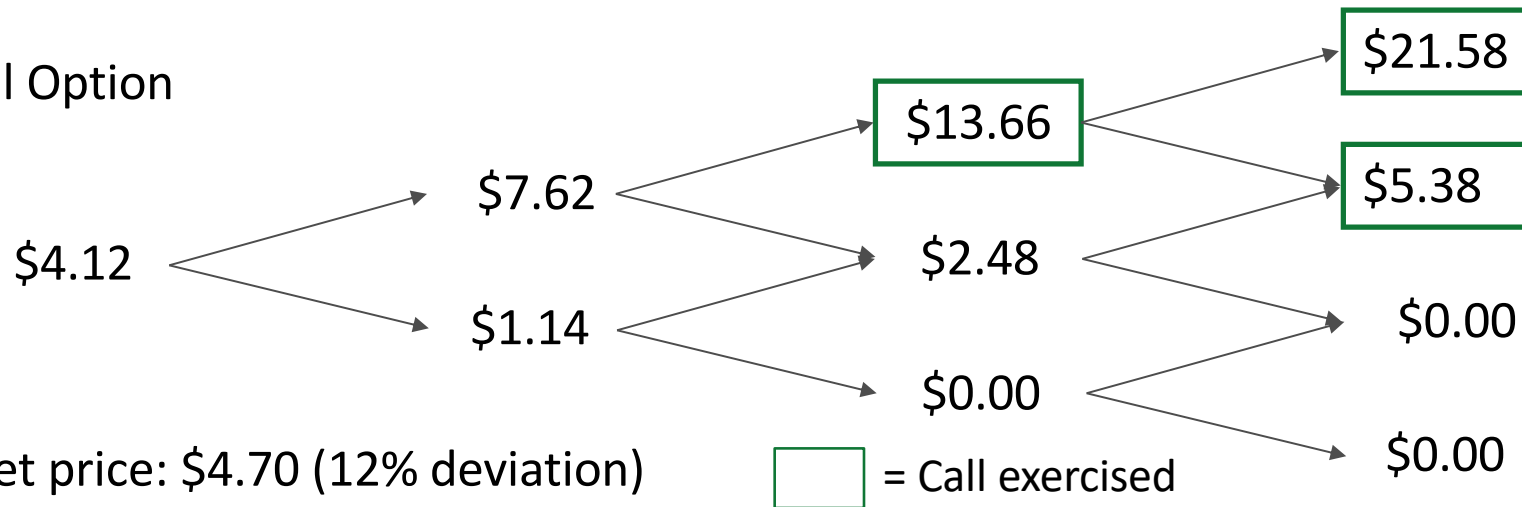
Binomial Tree Valuation of Listed Options + Volatility and Dividend Yield Calibration

Stock Price: 1-year ($\delta = 8.1\%$)

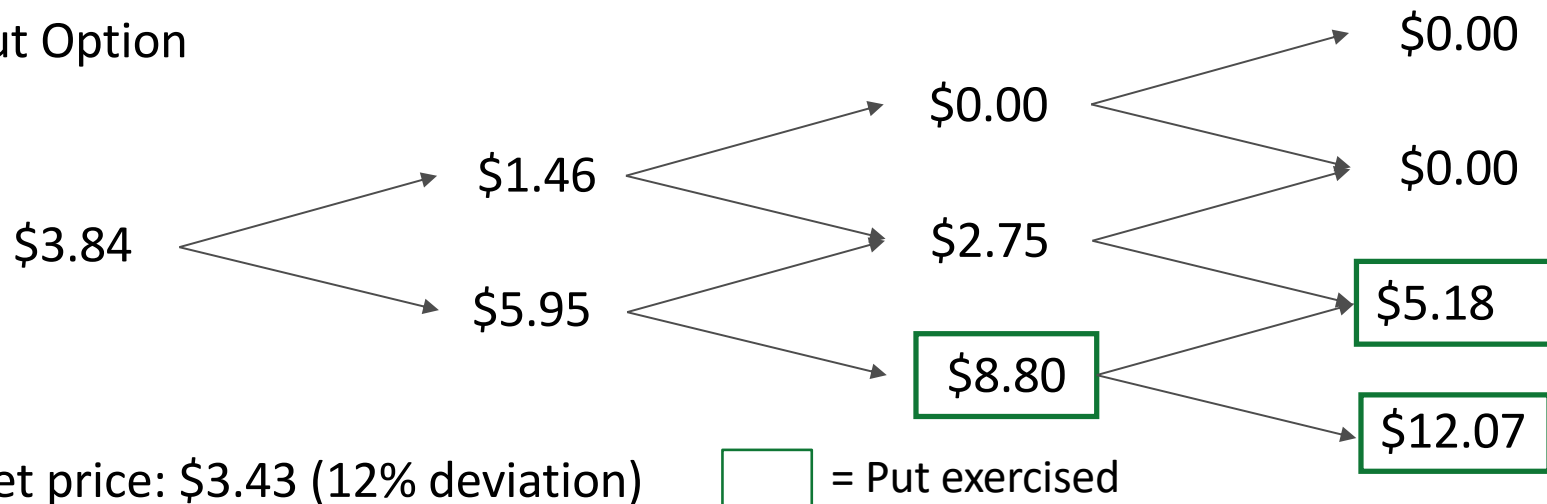


Listed Options: 1-Year, \$25 Strike ($\delta = 8.1\%$)

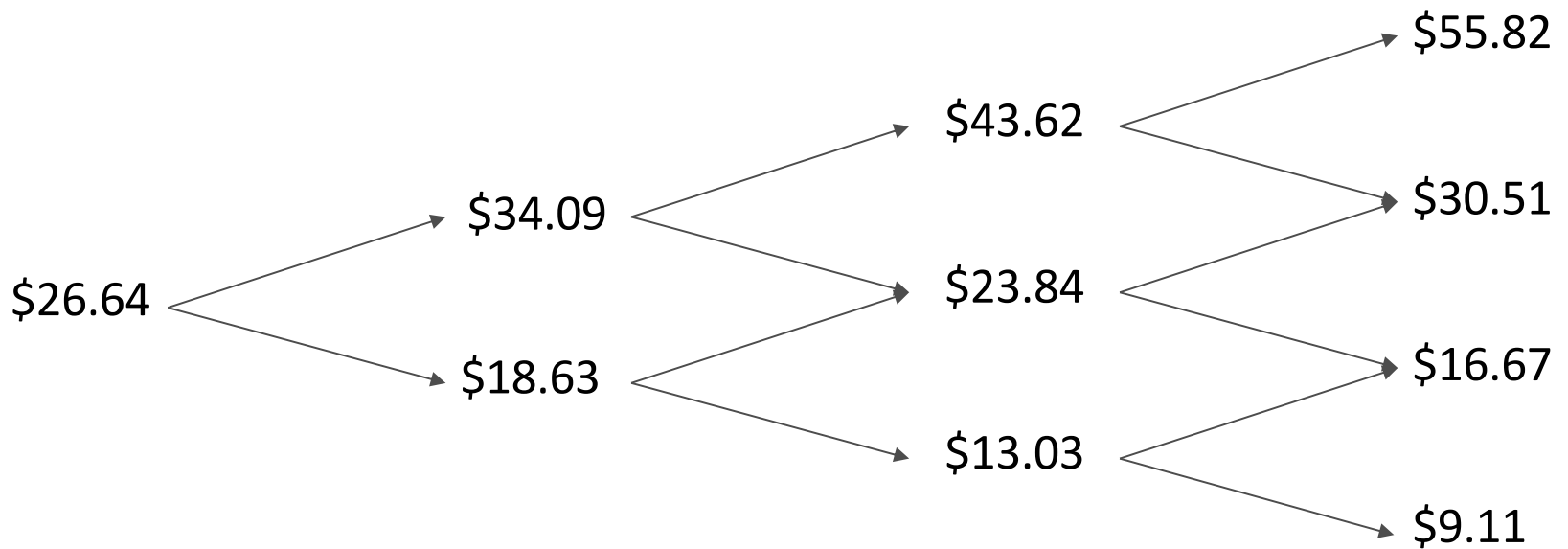
- Call Option



- Put Option

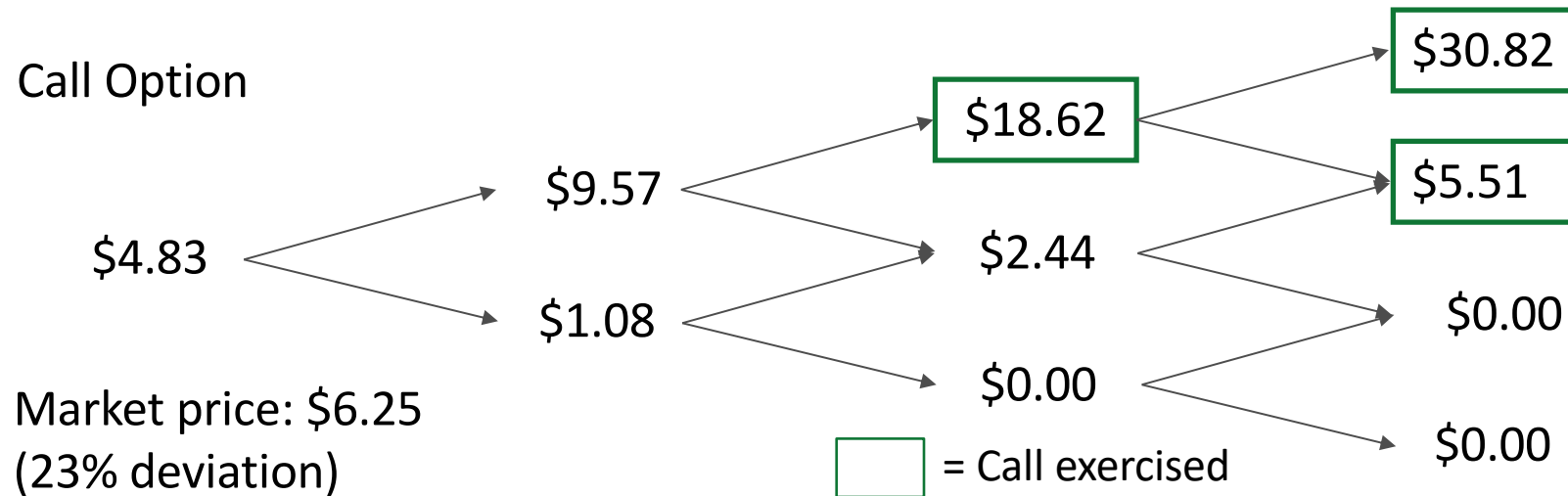


Stock Price: 2-year ($\delta = 8.1\%$)

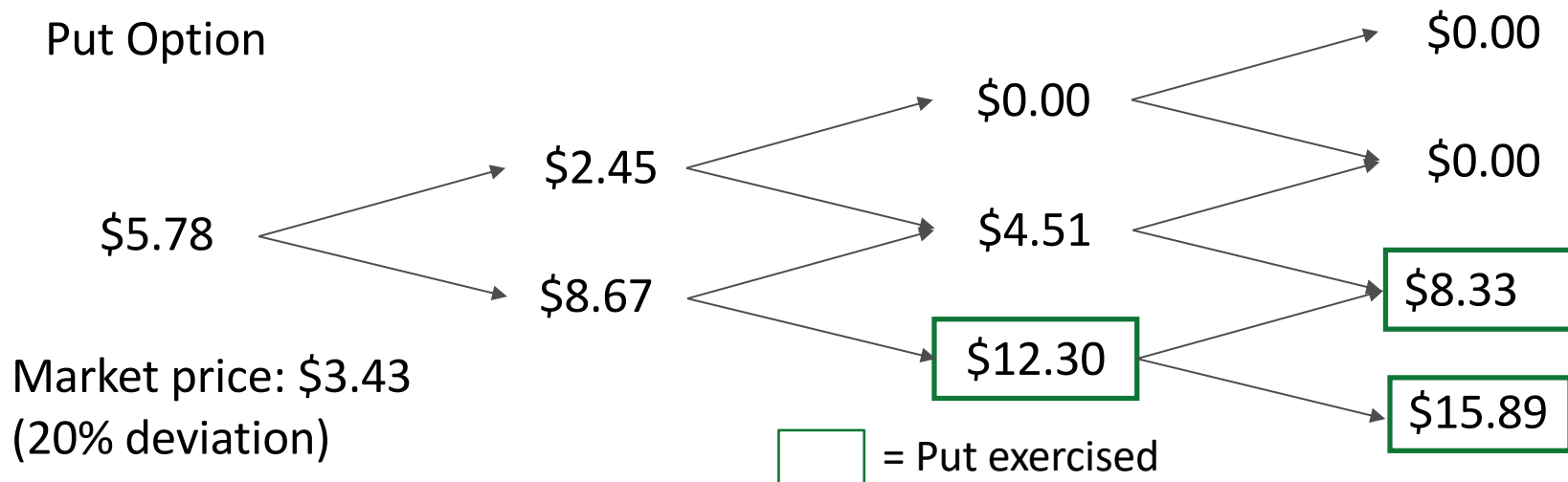


Listed Options: 2-Year, \$25 Strike ($\delta = 8.1\%$)

- Call Option



- Put Option



Listed Options: Recap

**No
dividend**

Maturity	Type	Strike	Mkt price	Binomial price	Deviation
1 year	Call	\$25	\$4.70	\$5.21	11%
	Call	\$30	\$2.61	\$3.03	16%
	Put	\$25	\$3.43	\$3.06	-11%
	Put	\$30	\$6.25	\$5.78	-8%
2 year	Call	\$25	\$6.25	\$7.08	13%
	Call	\$30	\$4.30	\$5.05	18%
	Put	\$25	\$4.83	\$4.45	-8%
	Put	\$30	\$7.73	\$7.22	-7%

**8.1%-dividend
yield (historical
rate)**

Maturity	Type	Strike	Mkt price	Binomial price	Deviation
1 year	Call	\$25	\$4.70	\$4.12	-12%
	Call	\$30	\$2.61	\$1.93	-26%
	Put	\$25	\$3.43	\$3.84	12%
	Put	\$30	\$6.25	\$6.54	5%
2 year	Call	\$25	\$6.25	\$4.83	-23%
	Call	\$30	\$4.30	\$2.78	-35%
	Put	\$25	\$4.83	\$5.78	20%
	Put	\$30	\$7.73	\$8.53	10%

What Happened to Citigroup's Dividend?



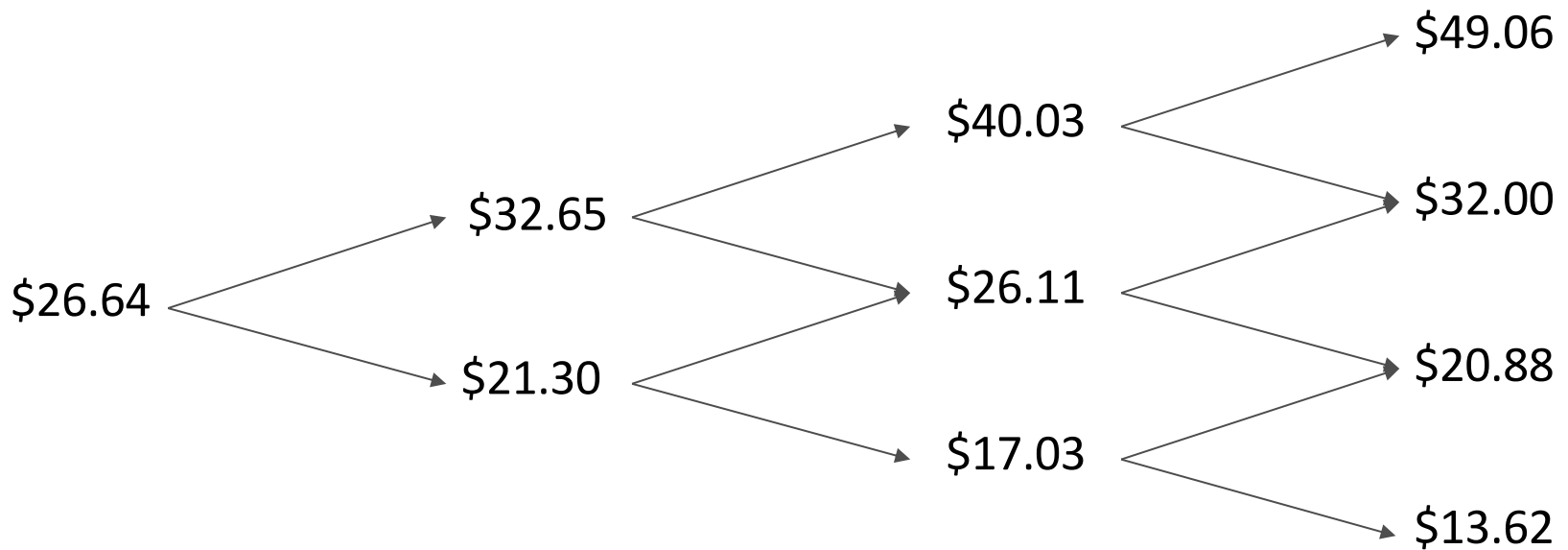
- 2007: 3rd-largest dividend payer in the S&P 500
- 2008: Bailed out by the U.S. government
- ➔ 2009-10: Must suspend dividends
- 2011: Resumed dividends but yield $\approx 0.10\%$

Volatility and Dividend Yield Calibration

Average Absolute Deviation from Actual Price Across All Traded Options												
Volatility	Dividend yield											
	0%	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
	30%	17%	19%	20%	22%	23%	23%	24%	24%	23%	25%	27%
	31%	14%	16%	17%	19%	20%	21%	21%	21%	22%	24%	26%
	32%	12%	13%	15%	16%	17%	18%	19%	19%	21%	23%	26%
	33%	11%	10%	12%	13%	14%	15%	16%	18%	19%	22%	25%
	34%	10%	8%	9%	10%	11%	12%	14%	16%	19%	22%	25%
	35%	11%	7%	6%	7%	8%	10%	12%	15%	18%	22%	25%
	36%	11%	7%	3.6%	4.0%	5.7%	8%	11%	14%	18%	21%	25%
	37%	11%	7%	3.5%	2.0%	3.9%	7%	11%	14%	18%	21%	24%
	38%	12%	8%	4.3%	2.1%	3.3%	7%	10%	14%	17%	21%	24%
	39%	12%	8%	6%	5%	4%	6%	10%	13%	17%	20%	24%
	40%	13%	10%	9%	8%	6%	7%	9%	13%	16%	20%	23%

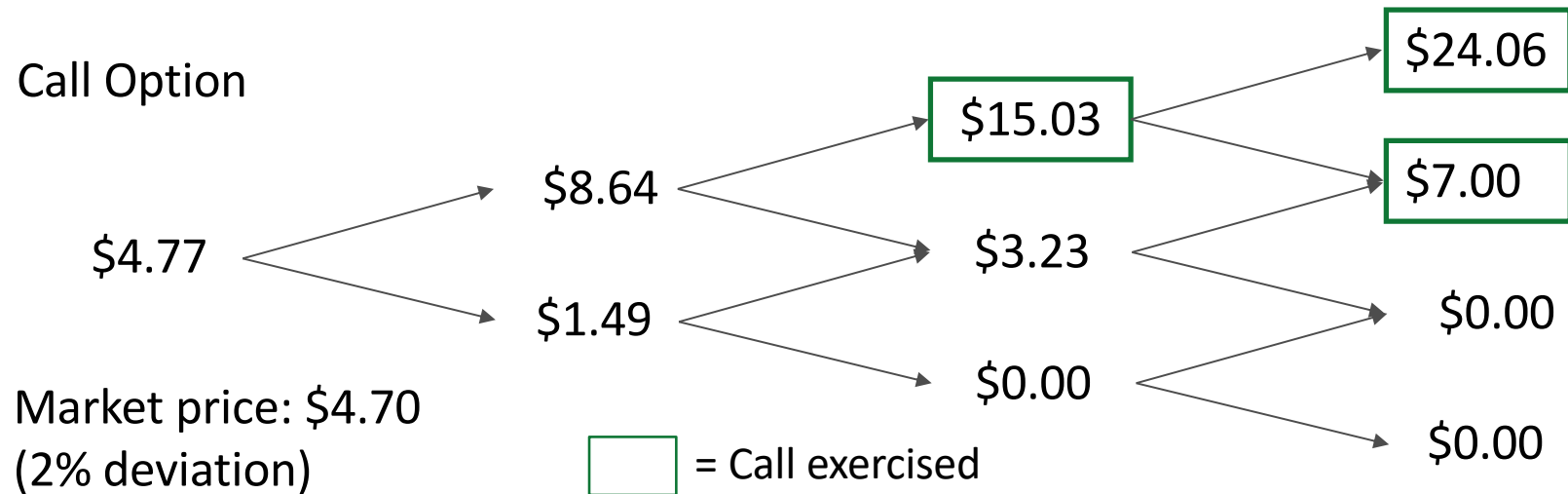
MAXIMUM ABSOLUTE DEVIATION FROM ACTUAL PRICE ACROSS ALL TRADED OPTIONS												
Volatility	Dividend yield											
	0%	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
	30%	33%	30%	29%	33%	38%	43%	49%	53%	55%	58%	61%
	31%	29%	27%	24%	29%	34%	39%	45%	50%	53%	55%	58%
	32%	26%	24%	21%	24%	29%	35%	42%	47%	50%	53%	56%
	33%	23%	21%	18%	19%	25%	32%	38%	44%	48%	50%	53%
	34%	20%	17%	15%	15%	22%	28%	34%	40%	45%	48%	51%
	35%	17%	14%	12%	11%	18%	24%	30%	36%	42%	45%	48%
	36%	14%	11%	9%	8%	14%	20%	26%	32%	38%	43%	46%
	37%	18%	10%	5%	4%	10%	16%	23%	29%	35%	40%	43%
	38%	21%	15%	9%	5%	8%	13%	19%	25%	31%	37%	41%
	39%	25%	19%	13%	8%	11%	15%	18%	22%	27%	33%	38%
	40%	30%	24%	18%	13%	14%	17%	21%	25%	28%	31%	36%

Stock Price: 1-year ($\delta = 3\%$)

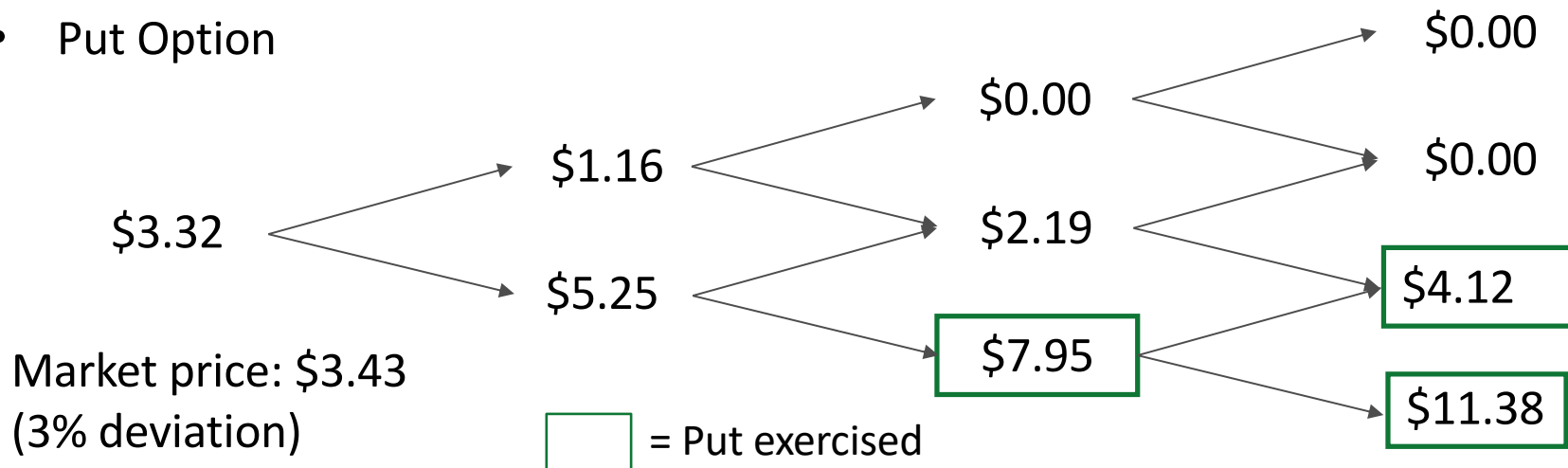


Listed Options: 1-Year, \$25 Strike ($\delta = 3\%$)

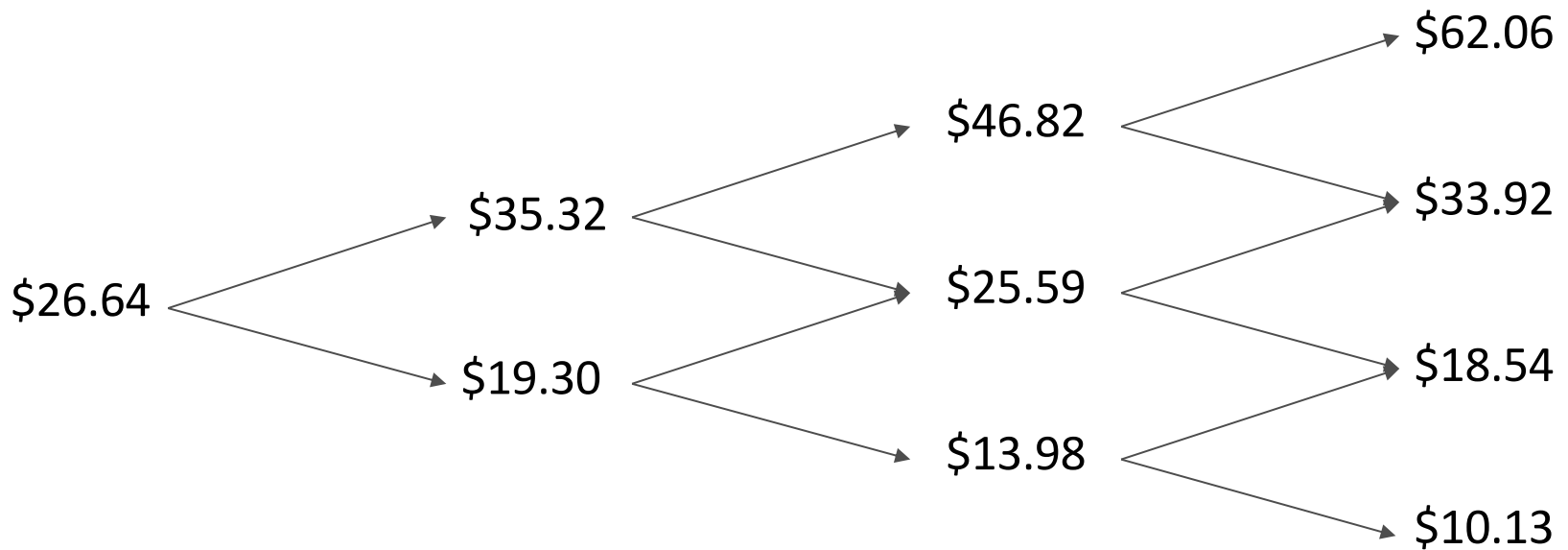
- Call Option



- Put Option

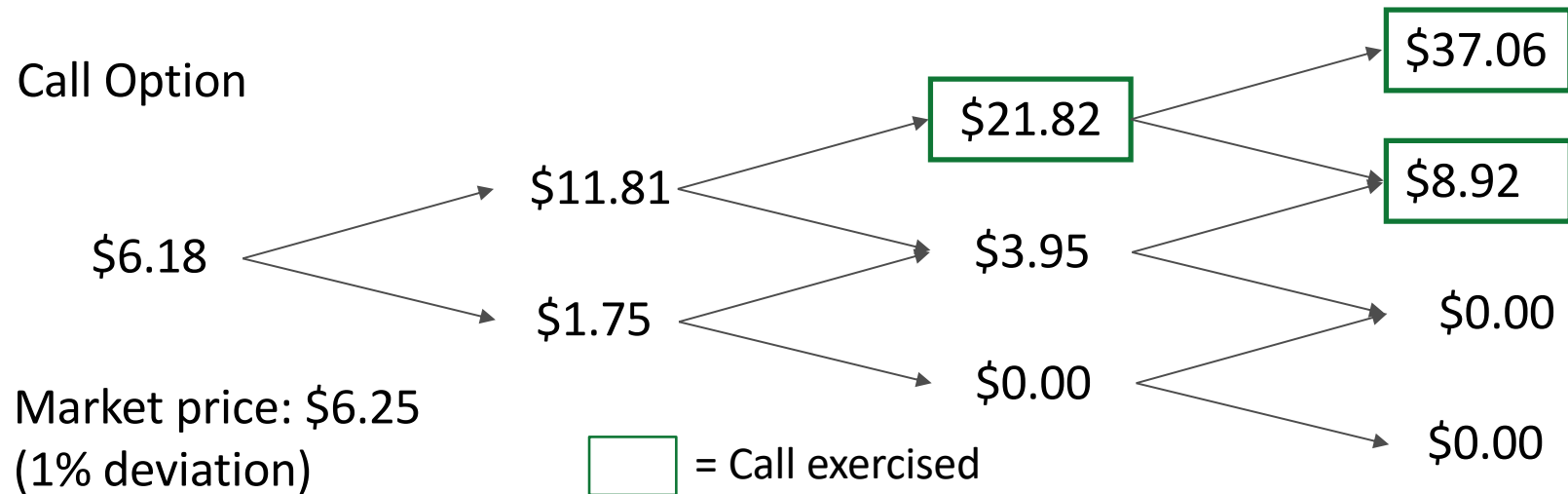


Stock Price: 2-year ($\delta = 3\%$)

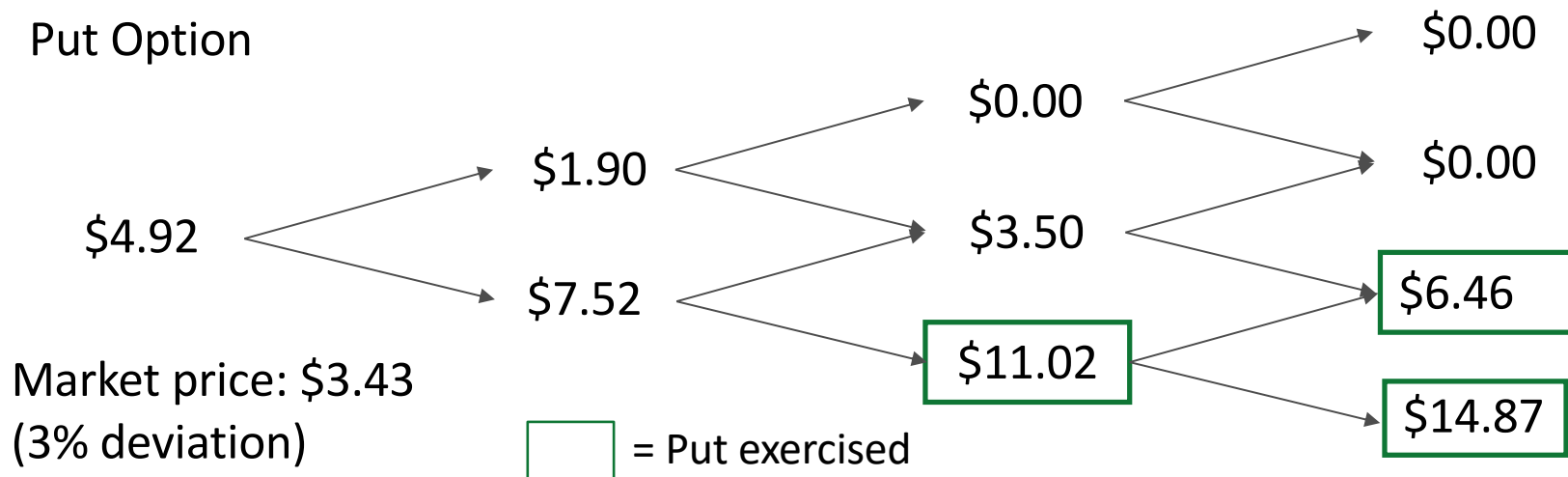


Listed Options: 2-Year, \$25 Strike ($\delta = 3\%$)

- Call Option



- Put Option



Dividend Calibration using Listed Options

3%-dividend yield (best match)

Maturity	Type	Strike	Mkt price	Binomial price	Deviation
1 year	Call	\$25	\$4.70	\$4.77	2%
	Call	\$30	\$2.61	\$2.58	-1%
	Put	\$25	\$3.43	\$3.32	-3%
	Put	\$30	\$6.25	\$6.04	-3%
2 year	Call	\$25	\$6.25	\$6.18	-1%
	Call	\$30	\$4.30	\$4.13	-4%
	Put	\$25	\$4.83	\$4.92	2%
	Put	\$30	\$7.73	\$7.69	0%